

Decoding divides: The role of socioeconomic status and personality traits in AI divides and educational inequality

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HIGHLIGHTS

- Digital literacy, not AI usage intensity, mediates personality-performance links.
- A new digital skills divide emerges, driven by personality traits instead of SES.
- SES advantages on performance operate independently of AI engagement.

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ABSTRACT

This study investigates whether the integration of AI in primary education relates to the digital divide and educational inequality, thereby examining the mediating roles of AI usage and digital literacy. Using survey and national registry data from 4497 Grade 6 students in the Netherlands (2022–2023), we examined relations between family socioeconomic status (SES) and personality traits (openness, extraversion, perseverance) via AI usage and digital literacy to academic performance. Our analyses revealed that digital literacy, not AI usage, positively mediated the relationship between all three personality traits and academic achievement. Family SES demonstrated a strong, direct relationship with performance but was not positively mediated by either AI usage or digital literacy, indicating that SES advantages persisted regardless of classroom AI implementation we observed. These findings suggest that as technology access becomes ubiquitous, a skill-based gap is evident that correlates with personality traits. This highlights the need for educational policies to move beyond providing mere access to AI and focus on cultivating digital skills tailored to diverse student personal attributes to ensure educational equality.

1. Introduction

In primary education, Artificial Intelligence (AI) has enabled adaptive learning that can address diverse students' needs with real-time, tailored interventions and feedback (Holmes & Tuomi, 2022; Zhai et al., 2021). If AI is deployed in a manner that is both inclusive (ensuring all students have access to AI tools) and equitable (providing personalized adaptations based on each student's unique features and abilities), AI could potentially reduce educational inequality by offering tailored learning experiences (Organisation for Economic Co-operation and Development, 2024). Systematic disparities in access to, usage of, and benefits from digital technologies (known as the digital divide

(Van Dijk, 2020)) represent a significant barrier to achieving this potential. The AI-driven digital divide extends this concept to AI-enabled tools, where greater complexity and rapid technological evolution raise thresholds of usage and skills required for effective utilization of the AI tools (Clemence et al., 2026). Evidence from digital divide research suggests the persistence of disparities in digital technology usage and skills among students across educational levels, resulting from societal inequalities in positional characteristics (e.g., SES) as well as personal attributes (e.g., personality traits) (Van Dijk, 2020). Recent studies indicate that these disparities have already manifested in the domain of AI, particularly following the introduction and rapid adoption of generative AI tools (Daepf & Counts, 2024). Such an AI-driven digital divide may

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mediate the relationships between SES, personality traits, and academic performance, potentially amplifying rather than mitigating educational inequalities. However, the extent to which AI introduction in primary education relates to educational inequality remains unclear and largely unexplored.

Therefore, we addressed this gap within the framework of digital divide theory in the current study. We investigated how family socioeconomic status (SES) and personality traits are related to AI usage and digital literacy, and how these two factors are associated with academic performance among primary school students.

1.1. The digital divide theory

The digital divide theory provides a comprehensive framework for understanding how societal inequalities manifest in the context of information and communication technologies (ICTs) (Van Dijk, 2020). At its core, the theory posits that inequalities in both positional societal factors (e.g., SES) and personal characteristics (e.g., personality traits) create a fundamental divide in how individuals engage with and benefit from digital technologies (Van Dijk, 2020). The divide is conceptualized across three levels: physical access (first level), skills and usage (second level), and outcomes (third level) (Lythreathis et al., 2022). Regarding skills and usage, a distinction can be made between general digital literacy versus the use of AI. In the current study, we focus on the second-level digital divide, examining AI usage intensity and digital literacy as key measures of usage and skills, respectively. We define AI usage intensity as a combination of the extent and frequency of students' engagement with AI learning tools, across different tool types and usage contexts. Digital literacy encompasses the ability to understand, integrate, communicate, and create information safely through digital technologies (UNESCO Institute for Statistics, 2018).

Typically, individuals with more advantaged positional features and favorable personal characteristics enjoy more extensive physical access to digital devices (DiMaggio et al., 2004), develop superior digital skills (Hargittai, 2002), engage in more effective usage (Van Deursen & Van Dijk, 2014a), and consequently reap more beneficial outcomes from digital technologies (Van Deursen & Helsper, 2015).

In primary education, the digital divide relates to students' learning experiences and academic achievement across all three levels (physical access; skills and usage; outcomes) (Tang et al., 2025). On the first level, although the divide in hardware access in primary school classrooms in developed economies is largely bridged over time (Hohlfeld et al., 2008, 2017), disparities remain in the quality and utilization of digital resources (Van De Werfhorst et al., 2022). Additionally, on the second level, there are substantial variations in primary school students' digital skills and their ability to effectively use educational technologies (Dolan, 2016; Hohlfeld et al., 2017). The variations in skills and usage result in the third-level divide, where differences in outcomes from using educational technologies are evident among students (Zhao et al., 2022). Effective use of educational technologies leads to enhanced learning and better preparation for future opportunities (Dolan, 2016). However, these benefits are unevenly distributed among primary school students, as reflected in their learning experiences and outcomes (Wang et al., 2024).

Despite extensive research on digital divides in education focusing on traditional ICTs (Dolan, 2016; Wang et al., 2024), the relationship between AI technologies and digital divides in education remains largely unexplored (Mirazchiyski, 2025; Tang et al., 2025). AI tools differ significantly from traditional ICTs in their complexity and rapid evolution (Seshia et al., 2022). The complexity and evolving nature of AI create a dual burden whereby students must develop competencies in technologies whose functional requirements are themselves subject to continuous change. This complexity and evolution of AI tools may thus create unique challenges in educational settings. Understanding AI-specific divides is crucial as they could significantly influence educational equity and outcomes.

1.2. The role of SES in the digital divide and educational inequality

SES stands as a crucial positional factor in digital divide theory, serving as a primary determinant of an individual's capacity to utilize and benefit from digital technologies (Van Dijk, 2020).

In primary education, at the first level of the digital divide, primary schools in higher SES areas and students from higher SES families have access to more advanced digital learning tools and tech-rich home environments (Dolan, 2016; Gonzalez-Betancor et al., 2021; Liu, 2021).

At the second level (level of interests in the current study), SES significantly shapes students' digital competencies and usage. Children from higher SES backgrounds often benefit from greater exposure to technology, more opportunities for guided learning, and higher levels of parental digital literacy, leading to more advanced digital skills (Hohlfeld et al., 2017; Van De Werfhorst et al., 2022). The skills advantage translates into distinct usage patterns. Higher SES students tend to engage in more sophisticated, creative, and collaborative digital learning activities, while their lower SES peers may be limited to more basic, leisure-oriented usage (Ren et al., 2022).

The third level of the digital divide manifests in learning outcomes, reflecting the cumulative impact of disparities in both access and skills. The combined effect of unequal access to technology (first level) and divergent digital competencies and usage patterns (second level) contributes to significantly different learning experiences and educational attainment among students from various SES backgrounds (Van De Werfhorst et al., 2022; Wang et al., 2024).

SES-based digital divides in primary education hold additional significance given the well-established relationship between SES and academic performance. Research across disciplines consistently shows a positive relationship between family SES and academic performance (Blanden et al., 2023).

Despite extensive research on both digital divides and SES-academic performance relationships, empirical evidence remains limited on how the introduction of AI in primary education relates to the established SES-performance patterns. There is a risk that AI learning tools could further widen the digital divide and exacerbate educational inequality. The benefits of AI tools in enhancing learning outcomes require proper usage, skills, and integration into broader learning strategies (Kabudi et al., 2021; Zhai et al., 2021). Students from lower SES backgrounds, who often lack the necessary skills and opportunities, may not fully utilize AI tools, potentially missing out on these benefits (Hannan & Eynon, 2025). As a result, educational inequalities could be exacerbated, as students from lower SES backgrounds may fall further behind compared to their higher SES counterparts who are better positioned to exploit AI's educational potential.

1.3. The role of personality traits in the digital divide and educational inequality

Within the digital divide framework, personal categorical differences (e.g., differences in personality traits) are also key origins of disparities in engagement with digital technologies (Van Dijk, 2020). Different personality traits, as a core aspect of personal categorical differences, may contribute considerably to digital divides at the second (usage and skills) and third (outcomes) levels (Scheerder et al., 2017).

Personality traits are enduring patterns of thoughts, feelings, and behaviors that distinguish individuals from one another. They are typically measured using models such as the "Big Five" (openness, conscientiousness, extraversion, agreeableness, and neuroticism) (Roberts & Yoon, 2022).

Studies among adult populations have shown that adults with higher openness, extraversion, and perseverance (a facet of conscientiousness in the Big Five model) are more willing and motivated to use digital technologies more frequently and effectively (Ali et al., 2020; Correa et al., 2010; Joshi et al., 2023; Maran et al., 2022). As students share fundamental psychological characteristics with adults, and there is no evidence indicating that other personality traits might correlate with AI

usage and digital skills, we focus our investigation on these three specific traits (openness, extraversion, and perseverance).

While the aforementioned research has established personality effects on technology usage patterns, it has largely overlooked how these traits are correlated with digital skills, the second key component of the second-level digital divide (Scheerder et al., 2017; Van Dijk, 2020). This research gap may exist because traditional ICT tools were often relatively simple and static. Mastering them typically required procedural learning rather than a strong individual disposition for continuous adaptation or critical thinking. In contrast, the rapidly evolving and generative nature of AI presents a different challenge (Clemence et al., 2026). It necessitates greater user autonomy for acquiring and updating knowledge and requires enhanced critical evaluation of AI-generated content (Clemence et al., 2026; Glickman & Sharot, 2025). These abilities (autonomy and critical evaluation) are core components of digital literacy (Bawden, 2008) and are closely linked to traits like openness, extraversion, and perseverance (Maran et al., 2022). The demanding skill set required by AI, together with its alignment with these specific personality traits, suggests a need to reassess how personality traits correlate with digital skills in the context of AI.

Mirroring the scarcity of research in adult populations, educational research also lacks empirical evidence connecting students' personality traits to digital skills and AI usage. Nonetheless, emerging evidence indicates that personality traits are critical in technology adoption and directly influence academic performance. One strand of research from educational economics demonstrates that students with higher levels of perseverance (Brandt et al., 2021), openness (Meyer et al., 2019), and extraversion (Meyer et al., 2023) tend to achieve better academic outcomes. A second strand of research from developmental psychology shows that personality traits also shape important aspects of technology adoption for younger populations. Research highlights the importance of openness and extraversion in children's innovation ability (Rawlings et al., 2022), and perseverance is crucial for sustained engagement with complex learning tasks that demand iterative problem-solving and adaptation (Duckworth et al., 2007), which are essential for engaging with AI.

In sum, research findings across disciplines among different population groups suggest that openness, extraversion, and perseverance could significantly influence students' adaptability, effective use of, and benefits from AI learning tools in education. Therefore, AI usage and skills could potentially mediate the relationship between these personality traits and academic achievement. It is then warranted to examine their relationships with students' usage of and skills in AI learning tools, and how the differences in usage and skills contribute to inequality in academic performance.

1.4. The present study

While existing research provides valuable insights into the digital divide and its implications for education, several critical gaps remain in our understanding of how AI relates to the divide. First, numerous studies have examined the digital divide in (primary) education, including many prompted by COVID-19 after 2020 (Liu, 2021; Wijayanti et al., 2024), but they predominantly focus on traditional ICT-driven divides rather than the AI-driven ones (Wang et al., 2024). AI, being more complex and rapidly evolving than conventional ICT, may create unique challenges such as inequitable access to advanced tools and resources, as well as continuous acquisition and updating of skills (Wang et al., 2025). Even in developed countries where the first-level divide (physical access to basic digital hardware and internet connection) is largely bridged (Eurostat, 2024), the second-level (skills and usage) and third-level (outcomes) divides are still prominent (Van De Werfhorst et al., 2022), especially concerning AI-related tools (Appel et al., 2025; Chatterji et al., 2025; Daepf & Counts, 2024). Understanding the second-level divide is particularly crucial as it forms the basis for comprehending the third-level divide in AI-enhanced education (Van Dijk, 2020). In other words,

disparities in AI skills and usage patterns may directly influence the inequalities in outcomes derived from AI technologies. However, research on AI skills and usage disparities in education remains limited, leaving the nature and extent of AI-driven educational inequalities unclear.

Second, beyond the digital divide phenomenon itself, how AI integration in education relates to educational inequalities remains ambiguous and under-explored, despite the potential advantages and relevance of AI in primary education (Holmes & Tuomi, 2022). Research has not established how the introduction of AI in primary education will alter the well-established relationship between family SES and academic outcomes (Blanden et al., 2023).

Third, there is a significant gap in our understanding of how personality traits relate to digital proficiency, particularly in the context of AI. In digital divide theory and empirical research on adult populations, there is a lack of studies exploring how personality traits predict digital skills, a crucial component of the second-level divide (Van Dijk, 2020). The research gap becomes even more pronounced in the field of education. While educational research has examined the second-level digital divide, educational studies have overlooked the role of personality traits in AI contexts, especially regarding AI learning tools (Mirazchiyski, 2025). The neglect of personality factors is particularly significant given the established importance of personality in predicting academic performance (Almlund et al., 2011). Understanding how personality traits associate with both the digital skills and AI usage in educational settings is crucial, as AI has the potential to amplify the effects of personality on educational outcomes through the second-level divide.

To address the identified research gaps, we investigate the following research question: To what extent do AI usage intensity and digital literacy mediate the relationship between family SES and academic performance on the one hand and between personality traits (openness, extraversion, and perseverance) and academic performance on the other hand?

Based on digital divide theory and existing empirical evidence, we hypothesize:

- H1: The relationship between family SES and academic performance is positively mediated by a) AI usage intensity and b) digital literacy.
- H2: The relationship between personality traits (openness, extraversion, perseverance) and academic performance is positively mediated by a) AI usage intensity and b) digital literacy.

To test these hypotheses, we employ a unique dataset combining survey responses and national administrative records from 4497 Grade 6 students in the Netherlands during the 2022–2023 academic year. We conduct mediation analyses using PROCESS macro Model 4 to examine whether AI usage and digital literacy mediate the relationships between individual characteristics (family SES and personality traits) and academic performance. The remaining sections of this paper are structured as follows: Section 2 describes the method, including the sample, procedure, and measures; Section 3 presents the results of the analyses; Section 4 discusses the findings and their implications; and Section 5 concludes.

2. Method

2.1. Participants

The sample was drawn from the 2022–2023 wave of the Education Monitor Limburg (in Dutch: Onderwijs Monitor Limburg, OML) survey. The OML targeted Grade 6 students ("group 8" in the Dutch education system, equivalent to U.S. 6th grade) in the province of Limburg. Limburg is a southern province in the Netherlands with lower average educational attainment and concentrated areas of additional governmental funding, providing substantial SES variation relevant for studying educational inequalities (Monfrance et al., 2025). Despite these regional differences, digital infrastructure in Limburg aligns with

the national standard, with near-universal classroom technology access achieved through national digitalization policies (Kerssens & Van Dijck, 2023). The institutional context is national, since Dutch educational policies, including curriculum standards and digital learning tool implementation, are set centrally and applied uniformly across provinces. The OML covers about 90% of regional schools, providing near-census coverage of the student population. Before excluding cases with missing data, the initial sample comprised 6772 students, representing approximately 75% of all Grade 6 students in the province that academic year. After excluding participants with missing data on key variables (the three personality traits, AI usage variables, digital literacy, and family SES), the final analytic sample consisted of 4497 students. Most missing observations lacked data on all variables, suggesting that missingness resulted from entire schools not participating rather than individual students skipping specific items. Because no data were collected from non-participating schools, a formal selectivity analysis comparing responding and non-responding institutions was not feasible.

The final sample was approximately balanced in terms of sex (52% male). The average age was 11.9 years ($SD = 0.47$).

2.2. Procedure

Data collection was conducted under a legally binding cooperation agreement between the participating school boards and Maastricht University, with the protocol approved by the local ethics committee (ERCIC-092-12-07-2018). Parents were informed about the study via written school communications with sufficient advance notice to exercise their right to withdraw their child. Passive consent was used following approval by the local ethics committee as appropriate for this minimal-risk educational survey research, and the withdrawal rate was negligible.

The OML survey was administered to students and teachers in-school under supervised conditions during the 2022–2023 academic year between March and June. The survey provided measures for personality traits, AI learning tool usage, digital literacy, and covariates including traditional information and communication technology in education (ICT-Ed) usage, IQ, oldest child status, age, and sex. Following data collection, the survey data were linked via a secure, encrypted identifier to national registry data incorporated by Netherlands Cohort Study on Education (Haelermans et al., 2020), to obtain students' primary school exit test scores and information on their parents' SES. The linkage

was performed by Statistics Netherlands (Centraal Bureau voor de Statistiek; CBS) using encrypted identifiers, and the linked dataset was pseudonymized. Access was restricted to authorized research team members via CBS Remote Access. The analysis was conducted via Remote Access on CBS secure local machine to ensure data security and privacy compliance.

2.3. Materials and measures

The following sections describe the measures used in this study. Descriptive statistics and correlations for all measures are presented in Table 1 in the Results section.

2.3.1. Registry-based measures

SES. SES was measured by a composite index derived from national tax registry data, combining 50% parental-average wealth percentile and 50% parental-average income percentile, where both components were ranked relative to the Dutch working population and the parental average was calculated as the mean of father's and mother's percentiles. As a robustness check, we re-specified SES using alternative measures including parental average wealth, parental average income, and highest parental education (see Appendix C.2 for details). These alternative specifications did not alter the substantive findings.

Academic performance. We measured academic performance using students' end-of-primary-education exit test scores from national registry data (2022–2023). Dutch law grants schools autonomy in selecting among approved test providers (Eurydice, 2025). Schools in our sample used five certified test providers with different scales and distributions, making raw scores non-comparable. Provider-specific raw-score distributions were heterogeneous and non-normal. We therefore converted scores to national within-provider percentiles, which aligned scales while preserving students' relative standing in each provider's scoring system. Percentiles were treated as continuous variables ranging from 0 to 100. Unlike z-score standardization, percentile conversions were more robust to skewness and heavy tails in the observed distributions. To absorb any remaining systematic differences between instruments, we controlled for test-provider identifiers in all models. As a robustness check, we also repeated all mediation analyses using z-score standardized outcomes; results were substantively unchanged (see Appendix C).

Table 1
Descriptive Statistics and Correlations for Grade 6 Students ($N = 4,497$).

Variable	<i>M (SD)</i>	1	2	3	4	5	6	7	8	9	10	11	12
1. Digital literacy	0.00 (0.92)	1.00											
2. AI usage intensity	0.21 (0.96)	0.22***	1.00										
3. Openness	−0.00 (0.88)	0.23***	−.00	1.00									
4. Extraversion	0.00 (0.82)	0.22***	−.04*	0.31***	1.00								
5. Perseverance	0.00 (0.92)	0.11***	0.03*	0.19***	0.15***	1.00							
6. Family SES	60.5 (20.5)	−.09***	0.01	0.10***	0.12***	0.08***	1.00						
7. ICT-Ed Usage	2.12 (1.33)	0.00	0.41***	0.02	−.01	0.03*	0.03	1.00					
8. IQ	−0.33 (0.82)	−.04**	0.01	0.16***	−.02	0.15***	0.16***	0.04**	1.00				
9. Oldest Child Status	0.49 (0.50)	0.05**	0.00	0.06***	0.03	0.02	−.05***	0.02	−.00	1.00			
10. Age	11.9 (0.47)	0.04*	0.02	−.08***	−.04*	−.04*	−.21***	0.01	−.11***	0.01	1.00		
11. Sex	0.52 (0.50)	−.26***	0.01	−.04**	−.08***	−.07***	0.03	0.06***	−.03	−.05**	1.00		
12. Academic score (percentile)	53.9 (27.1)	0.08***	0.00	0.23***	0.06***	0.26***	0.24***	0.23***	0.18***	0.01	−.03*	−.26***	1.00
13. Test provider: Cito	1374 (30.5%)	—	—	—	—	—	—	—	—	—	—	—	—
14. Test provider: Route 8	1674 (37.2%)	—	—	—	—	—	—	—	—	—	—	—	—
15. Test provider: IEP	758 (16.8%)	—	—	—	—	—	—	—	—	—	—	—	—
16. Test provider: AMN	426 (9.5%)	—	—	—	—	—	—	—	—	—	—	—	—
17. Test provider: Dia	267 (5.9%)	—	—	—	—	—	—	—	—	—	—	—	—

Note. All personality traits, digital literacy, AI usage intensity, and IQ are standardized factor scores. Correlations are Pearson's r (two-tailed), computed using listwise deletion; for dichotomous variables (Sex, Oldest child status), r is the point-biserial. ICT-Ed Usage was measured on a [1–5] Likert scale ([1 = never, 5 = very often]); Age is in years. Family SES is a composite index combining 50% parental-average wealth percentile + 50% parental-average income percentile (ranked relative to Dutch working population). Academic score is a percentile within each test provider; test provider identifiers (rows 13–17) are mutually exclusive; frequencies/percentages refer to the analytic sample; correlations are thus not applicable (—). *** $p < .001$, ** $p < .01$, * $p < .05$.

2.3.2. Survey-based measures

AI usage intensity. We define AI usage intensity as the extent of students' engagement with AI learning tools across different tool types, usage contexts (at home, at school, or both), and frequency of use (categorical options: multiple times per day, once per day, twice per week, once per week). We constructed this measure by combining information from two complementary sources. Teachers reported which AI tools were used in their classrooms and where students were assigned to use them (at school only, at home as homework, or both), creating a set of tool-context indicators such as 'Gynzy at school only' or 'Snappet at both school and home.' Students self-reported their overall frequency of digital educational tool usage, providing individual-level engagement data. To create a composite index capturing this multi-dimensional information from both teachers and students, we required a method that could handle both the extensive margin (variety of tool-context combinations) and the intensive margin (whether specific combinations, such as Gynzy at home, empirically contributed to actual usage frequency). We adopted CatBoost (Prokhorenkova et al., 2018), a gradient boosting algorithm specifically designed for categorical features, because our tool-context indicators are inherently categorical (tool type $\times$ context combinations), and standard methods like ordinary regression or principal components analysis would impose inappropriate assumptions about distances between categories. Unlike ordinary methods that treat categories as arbitrary numbers or require manual dummy coding, CatBoost handles categorical variables natively through ordered target statistics, preventing information loss and reducing bias. This supervised, data-driven weighting approach trained the model to predict students' self-reported frequency from their tool-context indicator profiles. The model's learned feature-importance values quantified how much each configuration contributed to observed engagement frequency. Critically, each student received their own AI usage intensity score, computed as the weighted sum of whichever tool-context indicators applied to that individual student (for instance, a student using Gynzy at school and Snappet at home would receive weights only for those two combinations). This approach captured both the extensive margin (variety of tools used) and the intensive margin (contexts that empirically mattered more for predicting actual engagement), yielding individual-level variation that reflected heterogeneous patterns of AI integration. Compared to simple equal-weight counts, this method better captured heterogeneous AI adoption patterns while remaining interpretable (see Appendix A.1 for full technical details). The complete questionnaire items for AI learning tools usage are provided in Appendix A.4. Fig. A.1 shows that the resulting index provided smoother, more differentiated variation than simple counts or coarse context categories, enabling more precise detection of individual differences in engagement that may relate to academic outcomes.

Digital literacy. We operationalized digital literacy using a Dutch translation of the PISA 2018 ICT Familiarity Questionnaire (nine items, see Appendix A.2 for the full questionnaire), which spans digital competency and digital autonomy (Organisation for Economic Co-operation and Development, 2019). All items were rated on a five-point Likert scale ranging from "strongly disagree" to "strongly agree."

Exploratory factor analysis indicated a single underlying dimension for the nine items, with the scree plot showing a dominant first eigenvalue (3.81). We therefore estimated a composite score of digital literacy for each student using a Graded Response Model (GRM), an Item Response Theory (IRT) model for ordered responses (Embretson & Reise, 2013). This approach yielded personal scores across all nine questions on a common scale, with IRT diagnostics indicating the highest measurement precision around the moderate ability range. Digital literacy scores were standardized to $M = 0$, $SD = 1$ for analysis. We used this composite measure rather than modeling subdimensions (e.g., operational, informational, critical) because the single-factor structure empirically supported aggregation.

Personality traits. We measured three personality traits relevant to educational and digital engagement: Perseverance, Extraversion, and Openness. They were assessed using Likert-type questions adapted from the Big Five Inventory-2 (BFI-2; Soto and John (2017)) and translated into Dutch. All items were rated on a five-point Likert scale ranging from "strongly disagree" to "strongly agree." Similar to digital literacy, we constructed a composite score for each trait using a GRM to utilize its advantage in modeling Likert-type data. All personality trait scores were standardized to $M = 0$, $SD = 1$ for analysis. The complete questionnaire items are provided in Section A.3.

Covariates. To account for potential confounding variables, we included several covariates. We measured IQ using the Non-Scholastic Cognitive Abilities Test (NSCCT), a standardized assessment of cognitive skills not tied to school curricula (Van Batenburg & Van der Werf, 2004). The results of the OML survey provided the NSCCT scores, which were derived from a 2-parameter logistic (2PL) Item Response Theory (IRT) model and standardized to $M = 0$, $SD = 1$. (Traditional) ICT-Ed Usage was assessed through a teacher-reported questionnaire on the frequency of using non-AI digital tools, such as publisher's digital resources and open-source software. This was operationalized as a sum score of the different tools used. Finally, demographic variables including Oldest Child Status (a binary variable), Sex, and Age were obtained from registry records by the OML project. Oldest Child Status and Sex were coded as binary variables (1 = yes and male, respectively), while Age was measured in years.

2.4. Analytic strategy

We tested our hypotheses using mediation analyses with the PROCESS macro 5.0 (Model 4) for R, an ordinary least squares (OLS) regression-based path analysis tool for estimating direct and indirect effects (Hayes, 2022).

Model specification. To test our mediation hypotheses, we specified a parallel mediation model examining the indirect relationships between four independent variables (family SES, openness, extraversion, and perseverance) and academic performance through two parallel mediators (AI usage intensity and digital literacy). We note that all mediation effects in the current study decompose correlational pathways and do not imply causation. The model follows Hayes' conceptual framework (Hayes, 2022) with three key pathways for each independent variable: (a) the relationship between the independent variable and each mediator (path a), (b) the relationship between each mediator and the dependent variable controlling for the independent variable (path b), and (c') the direct relationship between the independent variable and the dependent variable controlling for both mediators. The indirect effect through each mediator is calculated as the product of the respective paths a and b (i.e., $a \times b$). The model tests H1 by examining SES pathways through both digital factors, and H2 by examining personality trait pathways through both digital factors (see Fig. 1 for the complete model visualization).

Statistical approach. The model was estimated using ordinary least squares regression with robust standard errors clustered at the school level to account for potential within-school correlations in the data. All GRM-derived variables (digital literacy, openness, extraversion, perseverance, IQ) were on standardized scales ($M = 0$, $SD = 1$) by default. Missing data were handled using listwise deletion, resulting in a final analytic sample of 4497 students with complete data on all key variables. The school-level clustering of missing data does not compromise the validity of our findings given the substantial remaining sample size and broad school representation. Balance tests on participating versus non-participating schools conducted on the CBS secure local machine showed no significant differences in SES composition or academic performance, suggesting limited representativeness bias. The model included all control variables (student sex, age, IQ, traditional ICT-Ed

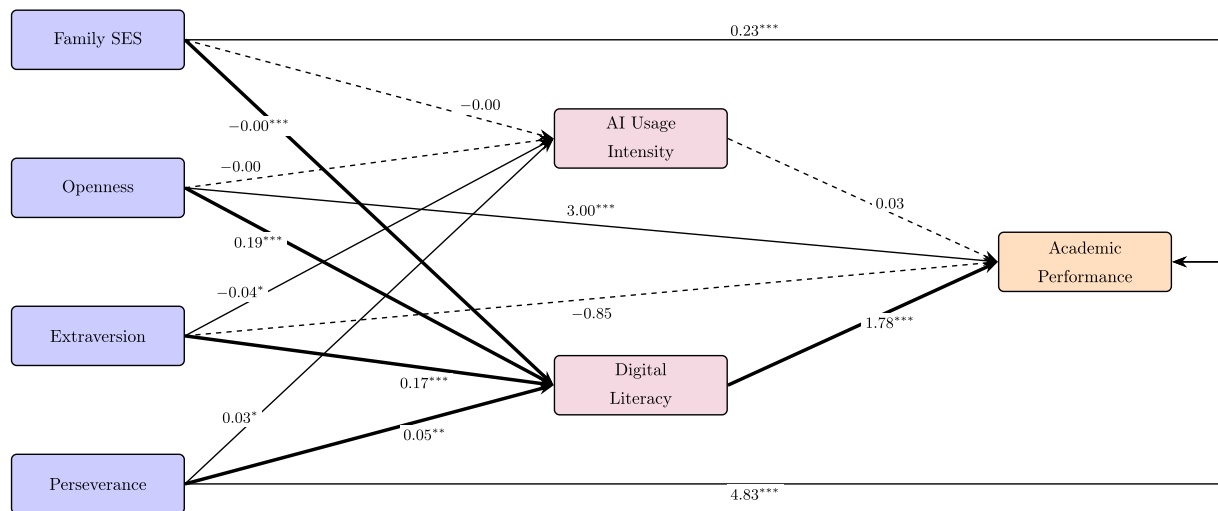


Fig. 1. Parallel Mediation Model: Family SES and Personality Traits Relating to Academic Performance Through AI Usage Intensity and Digital Literacy. Path coefficients are unstandardized ($N = 4,497$ Grade 6 students). Bold lines indicate significant indirect effects; dashed lines indicate non-significant paths. Significance levels: * $p < .05$, ** $p < .01$, *** $p < .001$. Digital literacy significantly mediated the relationships between personality traits and academic performance, while AI usage intensity showed no significant mediation effects. Detailed statistics and confidence intervals are reported in the main text and [Appendix B.1](#).

Usage, oldest child status, and test provider identifier) as covariates. For the mediation hypotheses (H1 and H2), we evaluated the significance of indirect effects using bias-corrected bootstrap confidence intervals with 5000 resamples (Preacher & Hayes, 2008). This approach is methodologically sound for nested data: cluster-robust standard errors correct for within-school correlation when testing path coefficients (Cameron & Miller, 2015; McNeish et al., 2017), while bootstrap confidence intervals provide superior inference for indirect effects, whose sampling distributions are typically non-normal (MacKinnon et al., 2004; Shrout & Bolger, 2002). Because our research question concerns population-average mediation effects rather than school-specific variance, this single-level approach with robust corrections is appropriate and explicitly supported by PROCESS version 5.0 (Hayes, 2022, 2025). An indirect effect was considered statistically significant if the 95% confidence interval excluded zero.

3. Results

Table 1 presents descriptive statistics for all continuous variables and correlation coefficients among key study variables. The sample demonstrated considerable variability in academic performance and SES.

The correlation analyses revealed several important patterns relevant to our hypotheses. Family SES showed significant positive correlations with the personality traits of openness and extraversion. Among the personality traits, openness and extraversion demonstrated the strongest significant correlations with digital literacy, followed by perseverance. Notably, AI usage intensity showed a strong significant positive correlation with traditional ICT-Ed Usage and a moderate positive correlation with digital literacy, but minimal correlations with personality traits and academic performance. Sex showed a significant negative correlation with digital literacy, meaning that female students tended to have higher digital literacy than male students.

3.1. Mediation analyses

Fig. 1 visually presents the complete mediation model results. The detailed statistical results for each path, including coefficients and confidence intervals, are documented in [Appendix B.1](#).

SES-related hypothesis (H1). For H1, family SES showed no significant mediation through either AI usage intensity or digital literacy. Family

SES demonstrated no significant relationship with AI usage intensity ($B < -0.0001$, $SE = 0.0003$, 95% $CI [-0.0006, 0.0006]$), and showed a statistically significant yet close-to-zero negative relationship with digital literacy ($B = -0.01$, $SE = 0.00$, 95% $CI [-0.01, -0.00]$). AI usage intensity showed no significant relationship with academic performance ($B = 0.03$, $SE = 0.66$, 95% $CI [-1.27, 1.33]$), while digital literacy showed a significantly positive relationship with academic performance ($B = 1.78$, $SE = 0.37$, 95% $CI [1.06, 2.51]$). These findings indicate that neither digital factor positively mediates the SES–performance relationship; therefore, H1 was not supported.

Despite the limited mediation effects, SES retained a strong direct correlation with academic performance in the mediation model ($B = 0.23$, $SE = 0.02$, 95% $CI [0.18, 0.27]$), suggesting that SES shows associations through channels beyond the measured digital factors.

Personality trait-related hypothesis (H2). The mediation analyses for personality traits provided partial support for our hypothesis, with notable differences across traits and mediators.

Across all three traits, the AI usage pathway yielded no mediation: AI usage intensity showed no significant relationship with academic performance ($B = 0.03$, $SE = 0.66$, 95% $CI [-1.27, 1.33]$), so indirect effects through this channel were negligible regardless of whether the trait–AI usage relationship was significant. The digital literacy pathway, by contrast, consistently mediated trait–performance relationships for all three traits, with digital literacy significantly relating to academic performance ($B = 1.78$, $SE = 0.37$, 95% $CI [1.06, 2.51]$) in each model.

For openness, the trait showed no significant relationship with AI usage intensity ($B = -0.00$, $SE = 0.02$, 95% $CI [-0.04, 0.04]$; indirect effect $B = -0.00$, 95% $CI [-0.01, 0.01]$) but was significantly related to digital literacy ($B = 0.19$, $SE = 0.02$, 95% $CI [0.15, 0.22]$), yielding a positive indirect effect ($B = 0.33$, 95% $CI [0.18, 0.50]$); with the direct effect remaining substantial ($B = 3.00$, 95% $CI [2.16, 3.83]$), the pattern indicates partial mediation. Extraversion showed a significant negative relationship with AI usage intensity ($B = -0.04$, $SE = 0.02$, 95% $CI [-0.07, -0.01]$; indirect effect $B = -0.00$, 95% $CI [-0.03, 0.03]$) while also being significantly related to digital literacy ($B = 0.17$, $SE = 0.02$, 95% $CI [0.13, 0.21]$) with a substantial indirect effect ($B = 0.30$, 95% $CI [0.16, 0.45]$); the direct effect of extraversion on academic performance was not significant ($B = -0.85$, 95% $CI [-1.73, 0.03]$), indicating that extraversion’s association with performance operates exclusively through digital literacy. Perseverance showed a significant

positive relationship with AI usage intensity ($B = 0.03$, $SE = 0.01$, 95% CI [0.01, 0.06]; indirect effect $B = 0.00$, 95% CI [-0.03, 0.03]) and was significantly related to digital literacy ($B = 0.05$, $SE = 0.01$, 95% CI [0.03, 0.07]) with a more modest indirect effect ($B = 0.08$, 95% CI [0.03, 0.16]); the direct effect remained large ($B = 4.83$, 95% CI [4.10, 5.57]), again indicating partial mediation.

In summary, digital literacy significantly yet partially mediated the relationships between all three personality traits (openness, extraversion, and perseverance) and academic performance, whereas AI usage intensity did not mediate any personality-academic performance relationships. For family SES, neither AI usage intensity nor digital literacy functioned as significant positive mediators in the SES-academic performance relationship, with SES showing a non-significant relationship with AI usage intensity and a statistically significant but practically negligible relationship with digital literacy.

To ensure the robustness of these findings, we conducted comprehensive sensitivity analyses that: (i) replaced the outcome with z-score standardized test scores (Appendix C.1); (ii) re-specified SES as income, wealth, and parental education (Appendix C.2); (iii) used a simple AI-tool count in place of the CatBoost-weighted index (Appendix C.3); (iv) tested alternative psychometric operationalizations for personality traits and digital literacy (Appendix C.4, Appendix C.5); (v) decomposed digital literacy into competency versus autonomy (Appendix C.6); and (vi) validated the pattern in a structural equation model (Appendix C.7). All findings remained consistent across the sensitivity analyses, with the only exceptions being the paths from extraversion and perseverance to AI usage intensity, which were not consistently significant across all specifications (see Appendix C for complete details; see Table 2 for a summary of all robustness checks).

4. Discussion

4.1. Summary of key findings

This study investigated how AI-related digital divides mediated the relationships between student characteristics and academic performance among 4497 Grade 6 primary school students in the Netherlands. Our findings align with the core tenets of digital divide theory while challenging some conventional assumptions and previous empirical results regarding the primary drivers of the digital divide in technology-rich contexts.

First, we examined the role of family SES. Contrary to our hypothesis, the results provided no support for the expected positive mediation of a) AI usage intensity and b) digital literacy between family SES and academic performance. H1 was not supported; neither AI usage intensity

nor digital literacy positively mediated the relationship between SES and academic performance. Instead, the direct effect of SES on academic performance remained strong and positive across all models ($B = 0.23$), indicating that socioeconomic advantages showed associations through pathways largely independent of digital literacy and AI usage among our sample.

Second, we investigated how personality traits are related to academic performance, finding partial support for our hypothesis. H2 was partially supported; digital literacy was a significant partial mediator for relationships between all three personality traits and academic performance. The mediation effects were particularly robust for extraversion (indirect effect = 0.30, 95% CI [0.16, 0.45]) and openness (indirect effect = 0.33, 95% CI [0.18, 0.50]), with a more modest but still significant effect for perseverance (indirect effect = 0.08, 95% CI [0.03, 0.16]). In contrast, AI usage intensity did not function as a mediator for any personality-performance relationship. Extraversion and perseverance were associated with AI usage intensity in opposite directions (higher extraversion was related to lower usage; higher perseverance was related to higher usage), but these differences did not relate to performance.

4.2. Interpretation of findings

The primacy of digital literacy over AI usage in predicting academic performance. Our finding that digital literacy significantly correlates with academic performance ($B = 1.78$) while AI usage intensity does not suggests that skills matter more than usage frequency in AI-enhanced education. In AI-integrated classrooms, AI exposure is routine; the real divide is not whether students use AI tools, but how effectively they do so. Our digital literacy measure captures AI-relevant competency and autonomy (comfort with new digital tools, problem solving, helping others, and the ability to find, install, and learn software independently), so it tracks who can convert AI exposure into academic gains. Even with a null AI-usage result, this can still be treated as an AI-related divide because the inequality lies in who can use AI effectively in AI-enriched classrooms. This result aligns with previous research (Hohlfeld et al., 2017; Wang et al., 2024) indicating that exposure to smart learning tools alone is insufficient for academic benefits. The pattern reinforces a fundamental aspect of digital divide theory: while bridging gaps in access and basic usage is necessary, it is far from sufficient for achieving digital equity (Van Deursen & Helsper, 2015; Van Deursen & Van Dijk, 2014a). The more consequential divide emerges in the realm of skills and effective use (Van Dijk, 2020), where students' underlying digital competencies (their abilities to critically evaluate, navigate, and employ digital technologies) prove more predictive of educational outcomes than mere usage intensity. As AI tools become more sophisticated and ubiquitous in educational settings (Chen et al., 2020), the ability to effectively navigate, evaluate, and utilize these technologies becomes a meta-skill that amplifies advantages in traditional academic domains (Su et al., 2023).

A new personality-associated digital skills divide. The SES-digital literacy relationship was statistically significant but practically negligible ($B = -0.00$), whereas the personality-digital literacy associations were positive and substantively larger (openness: $B = 0.19$, extraversion: $B = 0.17$, perseverance: $B = 0.05$). In this Dutch sample with high baseline digital access, personality traits thus showed stronger correlations with digital skills than SES background. Together with the significant and positive indirect effects through digital literacy (openness: $B = 0.33$, extraversion: $B = 0.30$, perseverance: $B = 0.08$), our results indicate that personality traits have emerged as important predictors of who develops the digital competencies needed for academic benefits in AI-enhanced environments.

Several explanations may account for why digital literacy mediates the relationship between personality traits and academic performance to varying degrees. The three personality traits (openness, extraversion, and perseverance) appear to relate to a proactive learning orientation and sustained engagement necessary to develop robust digital literacy

Table 2
Summary of robustness checks.

RC	Alternative specification	Details
1	Alternative Outcome Measures (z-score standardization of exit test score)	Appendix C.1
2	Alternative SES Measures (income, wealth, and parental education operationalizations)	Appendix C.2
3	Alternative AI Usage Measures (simple tool count vs. CatBoost-weighted index)	Appendix C.3
4	EFA-derived Personality Traits (EFA vs. GRM extraction methods)	Appendix C.4
5	EFA-derived Digital Literacy (EFA vs. GRM extraction methods)	Appendix C.5
6	Digital Literacy Components Analysis (digital competency and digital autonomy separately)	Appendix C.6
7	Structural Equation Modeling Validation (SEM vs. PROCESS macro)	Appendix C.7

Note. RC = Robustness Check; EFA = Exploratory Factor Analysis; GRM = Graded Response Model; SEM = Structural Equation Modeling. Robustness checks 1 to 6 maintain identical control variables, clustering procedures, and bootstrap methods as the main analysis.

in the context of using complicated AI learning tools. Students high in openness may be more willing to explore new digital environments and experiment with unfamiliar technologies (Kashdan et al., 2004). Those high in extraversion may be more likely to seek help, collaborate with peers, and engage actively with digital learning platforms (Borg et al., 2021). Students with greater perseverance may demonstrate the sustained effort required to master sophisticated digital skills that extend beyond basic operational competencies (Duckworth et al., 2007). These individual differences correlate with meaningful academic advantages through digital literacy (indirect effects: openness = 0.33, extraversion = 0.30, perseverance = 0.08). In a 25-student classroom ranked by openness, one standard deviation in openness (comparing a student at position 4 to one at position 13) translates to 0.33 percentile points higher academic performance through digital literacy alone (roughly 0.08 rank positions in a standard 25-student classroom). Openness and extraversion support digital literacy development more effectively than perseverance.

The findings reveal a pattern in the nature of digital divides in the context of AI-enhanced education. While digital divides have mostly been linked to disparities in “positional categorical inequalities” like SES (Scheerder et al., 2017; Van Dijk, 2020), our study provides empirical evidence that “personal categorical inequalities” such as personality traits serve as important predictors of the digital skills divide (Scheerder et al., 2017). To the best of our knowledge, this represents one of the first empirical investigations in educational research to show that personality traits mediate academic performance through digital literacy in AI-enhanced learning environments. Even in environments where technological access is equitable, substantial differences in digital engagement and benefits may persist based on personal attributes (Scheerder et al., 2017).

Non-mediation of SES-academic performance relationship. Our results showed that SES maintained strong direct effects on academic performance ($B = 0.23$) with no positive mediation through AI usage intensity ($B = 0.00$) or digital literacy (indirect $B = -0.01$). This finding contrasts with our expectations based on previous research (Zhang et al., 2024) and indicates that SES advantages on academic performance show associations through pathways beyond the classroom AI usage and digital literacy measures we observed.

In the context of a developed country like the Netherlands, where near-universal access to technology in schools has been largely achieved (Kerssens & Van Dijk, 2023), differences in SES background may no longer be the primary factor in whether and how children engage with digital technologies in educational settings. The provision of technology through schools and the institutional integration of AI tools may have effectively neutralized some of the traditional pathways through which SES relates to educational technology use. The result shows a relative success in addressing the SES-driven first-level divides (access and basic usage) and second-level divides (operational use of technology) up until the 2022–2023 school year.

The relationship between SES and digital literacy. The relationship between family SES and digital literacy was found to be statistically significant but practically negligible ($B = -0.00$, $SE = 0.00$, 95% $CI [-0.01, -0.00]$). This near-zero finding, which indicates that SES shows no meaningful relationship with digital literacy in our sample, contradicts the conventional expectation from digital divide literature that higher SES should correlate with higher digital literacy. This deviation from the expected positive relationship reflects two complementary mechanisms operating in the Dutch educational context. First, compensatory educational policies in the Netherlands provide schools serving lower-SES populations with additional resources for digital literacy through weighted student funding mechanisms (Kerssens & Van Dijk, 2023; Ladd & Fiske, 2011), actively reducing SES-based skill gaps. Second, higher-SES parents impose stricter limits on children’s recreational technology use (Livingstone & Helsper, 2008; Röhlke,

2025), paradoxically constraining informal digital skill development through unrestricted exploration that lower-SES children may gain through less-monitored access. Together, these mechanisms effectively level digital literacy across SES groups in this high-resource educational setting.

Differential effects of personality traits on AI usage patterns. Extraversion and perseverance were related to AI usage intensity in opposite directions (extraversion: $B = -0.04$, 95% $CI [-0.07, -0.01]$; perseverance: $B = 0.03$, 95% $CI [0.01, 0.06]$), though the resulting usage differences did not correlate with academic outcomes. In practical terms, more extraverted students used AI tools slightly less frequently, while more perseverant students used them slightly more, but neither pattern was correlated with their grades. These results might reflect that socially oriented students could favor peer- or teacher-mediated activities (Borg et al., 2021) over solitary interactions with AI tools, whereas perseverant students may sustain independent, task-focused engagement with AI tools (Duckworth et al., 2007) that aligns with classroom AI activities. The divergence of different personality traits’ effects on AI usage highlights the importance of designing AI-supported tasks that accommodate both collaborative use for socially oriented students and sustained, self-directed work for perseverant students, even though such usage variations do not directly impact achievement in our sample.

4.3. Implications of the study

Theoretical implications. This study makes two key contributions to digital divide theory in the context of AI-enhanced education. First, our findings demonstrate that digital divide theory remains highly applicable in the age of AI, with the second-level digital divide persisting even after first-level divides related to access have been largely addressed through policy interventions (Van De Werfhorst et al., 2022; Van Dijk, 2020). Bridging access gaps is insufficient; the more consequential divide emerges in skills and effective use (Van Deursen & Van Dijk, 2014b). Our results highlight that in AI-enhanced educational environments, individual differences in digital literacy continue to serve as critical mechanisms through which broader inequalities manifest.

Second, our findings reveal that personality-driven digital divides, representing personal categorical inequalities in the digital divide framework, deserve greater theoretical attention and empirical investigation than they have received in the past few decades (Van Dijk, 2020). While SES-initiated digital divides have dominated research and policy discourse over the past decades (Scheerder et al., 2017), our results suggest that personal characteristics, particularly personality traits, show equally important correlations with the second-level skills divide in the age of AI. The personality-driven pathway to digital inequality has been relatively overlooked in digital divide theory, despite its potential to create persistent educational advantages that operate independently of socioeconomic factors.

Practical and policy implications. The findings of this study carry significant implications for educational policy and practice. Our results suggest that policy efforts must fundamentally shift from simply providing AI tools and ensuring basic access toward cultivating the foundational digital skills that enable effective engagement with these technologies. This approach aligns with contemporary educational frameworks, such as the Dutch government’s comprehensive approach to basic skills (*basisvaardigheden*), which positions digital literacy alongside traditional competencies in language, mathematics, and citizenship as essential foundations for educational success (SLO, 2024), and European digital skills initiatives (European Commission, 2024).

The identification of personality traits as key predictors of digital skills carries two significant implications for educational interventions. First, it highlights the need to move beyond one-size-fits-all approaches and instead provide differentiated, heterogeneous support tailored to students’ differentiated personality profiles and individual learning approaches (Boekaerts, 1996; Borg et al., 2021). For instance, students

with high openness but low extraversion might thrive with self-directed discovery modules, while their more extraverted peers could benefit from collaborative projects (Kashdan et al., 2004; Rawlings et al., 2022). Second, beyond simply accommodating these differences, interventions can focus on actively cultivating these traits as malleable non-cognitive skills. While core personality is relatively stable (Almlund et al., 2011; Cunha et al., 2007), research demonstrates that targeted educational interventions can nurture the learning-related dispositions associated with them, such as perseverance (Alan et al., 2019; Roberts et al., 2017). Schools may consider using diagnostic assessments of students' openness, extraversion, and perseverance to inform targeted digital literacy development. While our study does not directly test specific interventions, our findings suggest that personality-responsive digital literacy instruction might be beneficial: students low in openness may benefit from guided exploration of digital literacy concepts with explicit rationales; those low in extraversion might respond better to small-group digital literacy training formats; those low in perseverance may need milestone-based digital skill development with incremental challenges. Teacher professional development focused on personality-responsive digital literacy pedagogy could help educators support students in developing the digital competencies that, as our findings show, mediate the relationship between personality and academic outcomes in AI-enhanced contexts.

Our findings also reveal important limitations regarding the scope of digital factors in addressing educational inequalities. The persistence of strong direct SES effects on academic performance, despite no positive mediation through AI usage or digital literacy, suggests that SES advantages continue to operate through pathways that are largely independent of the digital factors we measured. This finding indicates that digital literacy and AI usage interventions alone may be insufficient to close SES-based achievement gaps and educational inequalities. In contrast to these systemic SES-driven inequalities that require broader societal interventions, personality-driven digital divides offer more tractable targets for educational policy, as differentiated pedagogical approaches within existing school structures can directly address disparities in digital skills by tailoring support to individual personality profiles and cultivating learning-related dispositions.

4.4. Limitations

This study has several limitations that warrant consideration. First, the cross-sectional nature of our data prevents definitive causal conclusions. While our mediation analysis identified digital literacy as a key correlational pathway linking personality to achievement, the cross-sectional design does not establish temporal precedence. Alternative causal orderings, such as bidirectional relationships or common antecedents, cannot be ruled out. Longitudinal studies would be needed to assess developmental predictions, and causal effects require intervention studies. Second, our findings were drawn from Limburg, a single province in the Netherlands. Beyond the Netherlands, our findings, particularly the minimal SES to digital literacy relationship, likely reflect the country's relatively equitable digital infrastructure and mature school choice system (Eurostat, 2024; Kerssens & Van Dijk, 2023). Educational systems with greater technology access disparities or less developed digital infrastructure may observe stronger SES-driven digital divides than we detected. Third, our measures, while theoretically grounded, face certain constraints. The data were collected in Spring 2023, just before generative AI became widely integrated into educational contexts, meaning our findings on AI usage do not reflect the integration of generative AI tools into educational contexts (OpenAI, 2022). Furthermore, our reliance on teacher-reported AI usage likely captures institutionally sanctioned "technologies for education" but overlooks individual, out-of-school engagement with general-purpose "technologies in education" (Bauer et al., 2025). This measurement limitation is critical because the autonomous use of "technologies in education" may

represent the primary pathway through which new digital divides and educational inequalities emerge, yet our study provides no information on this potentially crucial dimension of AI engagement. Additionally, our AI usage measure relies on survey reports rather than objective behavioral data such as usage logs or system-generated activity records, which would provide more precise and unbiased measurements of actual engagement with AI tools. Our null AI usage finding may partly reflect this measurement constraint, and enhanced measurement approaches incorporating objective behavioral data might yield different results regarding AI usage effects. Using perseverance as a facet rather than the full conscientiousness dimension also limits direct comparability with studies that use complete Big Five inventories. Finally, because both personality traits and digital literacy were measured via student self-reports, the correlation between these constructs may be partially inflated by common-method bias, such as response styles or social desirability. Future research using performance-based digital literacy assessments or independent teacher ratings would mitigate this limitation and provide stronger evidence for the personality-digital literacy relationship.

Fourth, our study focuses exclusively on Grade 6 students in their final year of Dutch primary education and does not examine whether the relationships examined in this study vary across different developmental stages. Younger primary school students may show different patterns, where personality traits may be less stable and digital experience more limited, as may older secondary school students, where digital literacy patterns and educational contexts differ substantially. Future research should examine the personality-digital literacy-achievement relationships across age groups to understand how developmental stage moderates the patterns we observe.

4.5. Directions for future research

The limitations of this study highlight several important directions for future research. To address the issue of causality, longitudinal studies are needed to track the developmental trajectories of digital literacy and its relationships with personality and academic achievement over time. Experimental interventions could also be employed to test whether targeted pedagogical strategies can mitigate personality-based differences in digital engagement and skill development.

To enhance the generalizability of our findings, future research should include cross-national studies that examine how digital divides manifest across diverse educational and technological contexts. Furthermore, there is a pressing need to refine measurement approaches. Future work would benefit from incorporating behavioral measures or performance-based assessments of AI skills to strengthen validity (e.g., Jin et al., 2025). It is also crucial to develop more precise instruments capable of capturing individual-level AI engagement and differentiating between the use of institutionally provided "technologies for education" and general-purpose "technologies in education." A deeper understanding of the distinct inequality risks posed by each is essential for designing effective and equitable educational policies. Finally, as AI continues to evolve, ongoing research will be necessary to reconceptualize digital literacy and investigate the novel pathways through which individual characteristics shape learning outcomes in the digital age.

5. Conclusion

The integration of AI into primary education does not simply bridge old divides; it is associated with new ones, presenting a dual challenge to educational equity. Our findings reveal that as disparities in technological access narrow, the digital divide is re-emerging as a skills-based gap correlated with students' personality traits. For the first time, we provide empirical evidence that digital literacy serves as a key mechanism through which personality traits (openness, extraversion, and perseverance) correlate with academic advantages in AI-enhanced learning environments. This personality-driven divide is particularly concerning because it is rooted in stable individual differences, potentially creating

persistent inequalities that are harder to address than resource-based gaps.

Simultaneously, our results confirm that traditional socioeconomic divides persist with formidable strength. Family SES continues to demonstrate a strong direct correlation with academic performance, showing associations through channels beyond the digital factors we measured. This starkly demonstrates that AI is not a silver bullet for counteracting educational inequality. While AI transforms the mechanisms of digital disadvantage, it leaves the foundational pillars of socioeconomic inequality untouched.

These findings carry a clear mandate for policy and practice. Efforts to ensure equitable outcomes in a time of rapid AI development must adopt a two-pronged strategy: first, cultivating digital literacy through pedagogical approaches that are responsive to diverse student personalities, and second, redoubling efforts to dismantle the systemic socioeconomic barriers that digital advancements alone cannot erase.

CRedit authorship contribution statement

Zeyu Wang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sanne van Wetten:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Eliane Segers:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Carla Haelermans:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Anthropic's Large Language Model in order to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix Overview

Appendix A presents the methodological foundations for key measures, including the CatBoost construction of the AI usage intensity index and the complete survey instrument. **Appendix B** reports full mediation analysis and pathway decomposition results for the main PROCESS macro analysis. **Appendix C** documents seven robustness checks: alternative outcome operationalizations (z-scores); three SES re-specifications (income, wealth, parental education); an alternative AI usage measure (simple tool count); EFA-derived personality traits and digital literacy; a digital literacy component analysis (competency versus autonomy); and a structural equation model (SEM) replication of the main PROCESS macro results.

Appendix A. Methodological foundations and measurement construction

A.1. AI usage intensity construction: CatBoost weighting approach

Traditional measures of AI usage in educational settings typically rely on simple frequency counts or binary usage indicators, which fail to

capture the nuanced patterns of institutional AI integration and the differential importance of various tool-context combinations. For example, student A using multiple AI tools sporadically may not experience the same educational impact as student B who uses a single tool intensively in critical learning contexts. Yet, the binary or count-based measures treat these usage patterns equivalently.

The limitations on measurement are particularly problematic in heterogeneous educational environments where AI tools are deployed with varying intensity and contexts across different learning scenarios. This section details our machine learning approach to constructing a weighted composite measure that reflects both the extensive margin (variety of tools) and the intensive margin (contextual importance) of AI usage patterns. Our approach addresses the need for more sophisticated measurement instruments that can accurately distinguish between superficial AI exposure and meaningful AI engagement in educational contexts. The resulting metric provides a continuous measure that better captures individual differences in AI technology integration patterns compared to conventional counting approaches.

The construction of the AI usage intensity measure followed a two-step procedure. First, we implemented the CatBoost algorithm (Prokhorenkova et al., 2018) to derive the relationship between our predictor variables (context and count measures) and the outcome variable (students' overall frequency of digital tool usage).

Let X_i represent the vector of tool-context indicators for student i , and y_i denote their frequency of digital tool usage. The algorithm built an ensemble of decision trees through an iterative process:

$$F_M(X) = \sum_{m=1}^M h_m(X) \quad (A.1)$$

where $F_M(X)$ is the final prediction after M iterations, and $h_m(X)$ represents the m -th decision tree. At each iteration m , the model adds a new tree that minimizes the loss function:

$$L_m = \sum_{i=1}^n l(y_i, F_{m-1}(X_i) + h_m(X_i)) \quad (A.2)$$

where $l(\cdot)$ is the categorical cross-entropy loss function appropriate for our ordered categorical outcome.

Second, we extracted feature importance scores from the trained model. For each tool-context combination k , CatBoost calculated importance scores w_k by measuring the average reduction in prediction error attributed to splits on that feature across all trees:

$$w_k = \frac{1}{M} \sum_{m=1}^M \sum_{j \in S_{m,k}} \Delta L_{m,j} \quad (A.3)$$

where $S_{m,k}$ is the set of splits on feature k in tree m , and $\Delta L_{m,j}$ is the reduction in loss function achieved by split j .

Using these importance scores as weights, we constructed the AI usage intensity measure for student i as:

$$Intensity_i = \sum_{k=1}^K w_k X_{ik} \quad (A.4)$$

where X_{ik} is student i 's usage indicator (0 or 1) for tool-context combination k , and K is the total number of tool-context combinations. This weighted sum provided a continuous measure that captured both the extensive (variety of tools used) and intensive (importance-weighted contexts) margins of AI tool usage.

Fig. A.1 illustrates the distributions of three AI usage metrics in our sample: the simple count of AI tools used (ranging from 0–5 tools), the usage contexts (not available, school only, home only, or multiple contexts), and our constructed AI usage intensity measure. The bottom panel shows the distribution of our continuous AI usage intensity

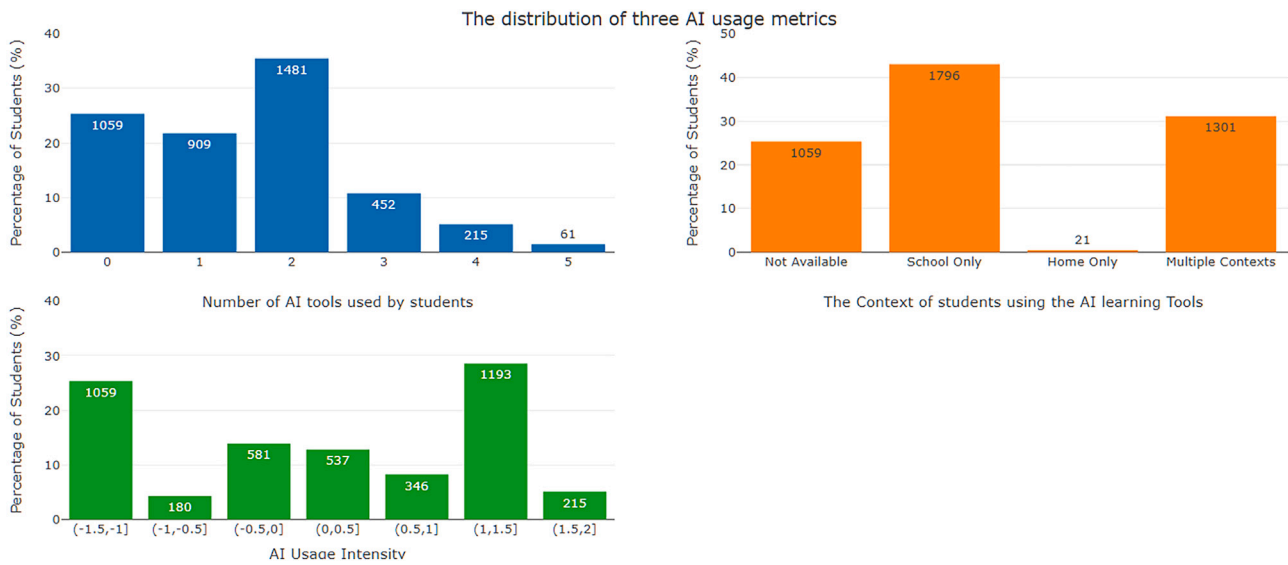


Fig. A.1. Distribution of AI Usage Metrics. The figure displays three complementary measures of AI engagement: (top left) number of distinct AI tools used by students, ranging from 0 to 5; (top right) contexts where students used AI learning tools; and (bottom) our constructed AI usage intensity measure that combines tool variety and contextual importance through CatBoost weighting.

variable, which ranged from approximately -1.5 to 2 , with the majority of students (26.5%) falling in the moderate usage range (1.1 – 1.5). This demonstrates the heterogeneity in AI engagement patterns that our weighted measure captures more precisely than simple tool counts.

To test the robustness of our findings, sensitivity analyses using a simple AI-usage count yielded the same substantive conclusions as our main analysis which used the AI usage intensity measure; see [Appendix C.3](#) for details.

A.2. Digital literacy questionnaire items

Digital literacy was measured using a Dutch translation of the PISA 2018 ICT Familiarity Questionnaire ([Kastberg et al., 2021](#)), comprising 9 items that assess two key dimensions: digital competency and digital autonomy. All items used a 5-point Likert scale with responses ranging from 1 = *helemaal niet* (not at all) to 5 = *helemaal wel* (completely), with all items positively coded so that larger scores indicate higher digital literacy. Each item is listed below in its original Dutch form, followed by an English translation in parentheses.

Items 1–6 assess digital competency:

- Ik werk graag met digitale leermiddelen (I enjoy working with digital learning tools).
- Ik heb er geen problemen mee om met nieuwe digitale apparaten te werken die ik nog niet ken (I have no problems working with new digital devices that I don't know yet).
- Als vrienden of familie nieuwe apparaten of apps willen kopen geef ik altijd advies (When friends or family want to buy new devices or apps, I always give advice).
- Ik kan goed overweg met de digitale apparaten thuis (I can handle the digital devices at home well).
- Als ik een probleem tegenkom met een van mijn digitale apparaten dan kan ik dat zelf oplossen (If I encounter a problem with one of my digital devices, I can solve it myself).
- Als anderen problemen hebben met hun digitale apparaten dan kan ik hen hiermee helpen (If others have problems with their digital devices, I can help them with this).

Items 7–9 assess digital autonomy:

- Als ik nieuwe software of apps nodig heb dan zoek ik zelf uit welke ik nodig heb (If I need new software or apps, I find out for myself which ones I need).
- Als ik nieuwe software of apps nodig heb installeer ik dat zelf (If I need new software or apps, I install them myself).
- Ik zoek en lees extra informatie over digitale apparaten zodat ik niet afhankelijk ben van anderen (I search for and read additional information about digital devices so that I am not dependent on others).

A.3. Personality trait questionnaire items

Personality traits were measured using questionnaires adapted from the Big Five Inventory-2 (BFI-2) ([Soto & John, 2017](#)) and translated into Dutch. Three traits relevant to educational and digital engagement were assessed: Openness, Extraversion, and Perseverance. All items used 5-point Likert scales, with response options specific to each scale as noted below. Each item is listed in its original Dutch form, followed by an English translation in parentheses; reverse-coded items are marked accordingly.

A.3.1. Openness scale

The five response options were: 1 = *helemaal niet* (not at all), 2 = *niet* (not really), 3 = *soms/weet ik niet* (sometimes/don't know), 4 = *wel* (yes), and 5 = *helemaal wel* (completely). All four items are positively coded:

- Ik barst van de ideeën (I am bursting with ideas).
- Ik heb een grote woordenschat (woordenschat is het totaal aantal woorden dat je kent) (I have a large vocabulary [vocabulary is the total number of words you know]).
- Ik ben geïnteresseerd in moeilijke ideeën (I am interested in difficult ideas).
- Ik heb heel goede ideeën (I have very good ideas).

A.3.2. Extraversion scale

The five response options were: 1 = *helemaal niet* (not at all), 2 = *niet* (not really), 3 = *soms/weet ik niet* (sometimes/don't know), 4 = *wel* (yes), and 5 = *helemaal wel* (completely). Items 1 and 2 are reverse-coded, so that higher scores on those items indicate lower extraversion:

1. Ik praat weinig / Ik praat niet veel (*reverse-coded*: I talk little / I don't talk much).
2. Ik ben stil in een groep van vreemden (*reverse-coded*: I am quiet in a group of strangers).
3. Ik sta graag in de belangstelling / ik vind het leuk om het middelpunt van de belangstelling te zijn (I like to be the center of attention / I enjoy being the center of attention).
4. Ik begin gesprekken met anderen (I start conversations with others).
5. Ik praat met veel verschillende kinderen op feesten (I talk to many different children at parties).

A.3.3. Perseverance scale

The five response options were: 1 = *helemaal niet* (not at all), 2 = *niet* (not really), 3 = *soms wel/soms niet* (sometimes/sometimes not), 4 = *wel* (yes), and 5 = *helemaal wel* (completely). Items 1, 2, and 3 are reverse-coded, so that higher scores on those items indicate lower perseverance:

1. Ik stop snel als iets te moeilijk wordt (*reverse-coded*: I quit quickly when something becomes too difficult).
2. Ik stop met een opdracht als het moeilijk is (*reverse-coded*: I stop with an assignment when it is difficult).
3. Als iets tegenvalt verlies ik gauw moed (*reverse-coded*: When something disappoints me, I quickly lose courage).

One additional item ("Ik geef niet op als ik het even niet meer weet"; I don't give up when I don't know what to do for a moment) was initially considered but excluded from the final scale due to poor psychometric performance in the IRT analysis. All reverse-coded items were appropriately recoded during analysis so that higher scores consistently indicate higher levels of the respective trait. The Graded Response Model (GRM) used for creating composite scores accounts for the direction of item coding in the analysis. In total, 5 out of 22 items across all scales required reverse coding: 2 extraversion items and 3 perseverance items.

A.4. AI learning tools usage questionnaire items

AI learning tools usage was assessed through teacher questionnaires measuring three key dimensions: which AI learning tools their students used, where students used them, and the frequency of student use. The questionnaire covered seven core AI learning platforms commonly used in Dutch primary education. All items were administered in Dutch to teachers during the 2022–2023 academic year. Each item is listed below in its original Dutch form, followed by an English translation in parentheses.

Teachers were asked to indicate which AI learning tools their students used, with a binary response format (1 = yes) for each tool:

1. Digitale adaptieve oefenprogrammas: Gynzy (AI learning tool: Gynzy).
2. Digitale adaptieve oefenprogrammas: Flexi (AI learning tool: Flexi).
3. Digitale adaptieve oefenprogrammas: Snappet (AI learning tool: Snappet).

4. Digitale adaptieve oefenprogrammas: Bingel (AI learning tool: Bingel).
5. Digitale adaptieve oefenprogrammas: Rekentuin (AI learning tool: Rekentuin).
6. Digitale adaptieve oefenprogrammas: Taalzee (AI learning tool: Taalzee).
7. Digitale adaptieve oefenprogrammas: Nieuwsbegrip (AI learning tool: Nieuwsbegrip).

For each tool used, teachers specified the usage context by responding to "Waar gebruik je [tool name]?" (Where do you use [tool name]?), selecting from three options: 1 = Op school (At school), 2 = Thuis als huiswerk (At home as homework), 3 = Op school en thuis (Both at school and at home).

Finally, students reported their overall frequency of digital learning tool usage by responding to: "Hoe vaak per week werken leerlingen op school met digitale leermiddelen?" (How often per week do students work with digital learning tools at school?). Response options were: 1 = meerdere keren per dag (multiple times per day), 2 = 1 keer per dag (once per day), 3 = 2 keer per week (twice per week), 4 = 1 keer per week (once per week), 5 = anders, namelijk... (other, namely...).

Appendix B. Complete mediation analysis results and pathway decomposition

B.1. Specific mediation results

The mediation analyses presented in the main text examined parallel mediation models (PROCESS Model 4) to test our hypotheses about the indirect relationships between independent variables (family SES and personality traits) and academic performance through AI usage intensity and digital literacy simultaneously. For each of the four independent variables (family SES, openness, extraversion, and perseverance), we estimated a parallel mediation model where both mediators (AI usage intensity and digital literacy) were included in the same model. This section provides comprehensive statistical details for all mediation pathways, including the direct relationships between independent variables and each mediator (path a), the relationships between each mediator and academic performance while controlling for the independent variable and both mediators (path b), the specific indirect effects through each mediator (calculated as the product $a \times b$), and the direct effects of independent variables on academic performance while controlling for both mediators simultaneously (path c').

The results in Table B.1 reveal the specific mechanisms through which individual characteristics correlate with academic outcomes in AI-enhanced educational environments. While AI usage intensity failed to serve as a significant mediator in any of the tested models, digital literacy emerged as a robust mediator for all three personality traits (openness, extraversion, and perseverance). The table demonstrates that despite some relationships between predictors and AI usage intensity, AI usage did not mediate the relationship with academic achievement for any predictor. In contrast, digital literacy showed significant positive mediation for all three personality traits but a statistically significant yet practically negligible mediation for family SES. These comprehensive results support our main findings that personality-driven differences in digital competencies, rather than simple AI tool usage or socioeconomic access to technology, represent the critical pathway through which individual differences translate into educational advantages in contemporary AI-enhanced learning environments.

Table B.1

Summary of mediation analyses predicting academic performance.

Model ($X \rightarrow M \rightarrow Y$)	Path a: $X \rightarrow M$		Path b: $M \rightarrow Y$		Indirect Effect ($a*b$)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage Intensity								
SES $\rightarrow$ AI Usage $\rightarrow$ Performance	-.0002 (.0013)	[-.003, 0.002]	0.029 (.66)	[-1.27, 1.33]	-.0000 (.0003)	[-.0006, 0.0006]	0.23 (.02)	[.18, 0.27]
Openness $\rightarrow$ AI Usage $\rightarrow$ Performance	-.004 (.022)	[-.048, 0.040]	0.029 (.66)	[-1.27, 1.33]	-.0001 (.006)	[-.015, 0.013]	3.00 (.42)	[2.16, 3.83]
Extraversion $\rightarrow$ AI Usage $\rightarrow$ Performance	-.037 (.015)	[-.067, -.007]	0.029 (.66)	[-1.27, 1.33]	-.0011 (.016)	[-.034, 0.032]	-.85 (.45)	[-1.73, 0.03]
Perseverance $\rightarrow$ AI Usage $\rightarrow$ Performance	0.034 (.015)	[.005, 0.063]	0.029 (.66)	[-1.27, 1.33]	0.0010 (.014)	[-.029, 0.030]	4.83 (.37)	[4.10, 5.57]
Mediator: Digital Literacy								
SES $\rightarrow$ Digital Literacy $\rightarrow$ Performance	-.0049 (.0007)	[-.006, -.004]	1.78 (.37)	[1.06, 2.51]	-.0088 (.002)	[-.013, -.005]	0.23 (.02)	[.18, 0.27]
Openness $\rightarrow$ Digital Literacy $\rightarrow$ Performance	0.187 (.018)	[.15, 0.22]	1.78 (.37)	[1.06, 2.51]	0.33 (.08)	[.18, 0.50]	3.00 (.42)	[2.16, 3.83]
Extraversion $\rightarrow$ Digital Literacy $\rightarrow$ Performance	0.168 (.017)	[.13, 0.20]	1.78 (.37)	[1.06, 2.51]	0.30 (.07)	[.16, 0.45]	-.85 (.45)	[-1.73, 0.03]
Perseverance $\rightarrow$ Digital Literacy $\rightarrow$ Performance	0.048 (.015)	[.02, 0.08]	1.78 (.37)	[1.06, 2.51]	0.085 (.03)	[.03, 0.16]	4.83 (.37)	[4.10, 5.57]

Note. $N = 4,497$. Results are from parallel mediation analyses (PROCESS Model 4) where both mediators (AI Usage Intensity and Digital Literacy) are included simultaneously in each model. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. For each model, we included the remaining independent variables (the other personality traits and SES) as covariates. Path b and direct effect (c') coefficients are the same for a given independent variable across both mediator rows because both mediators are estimated simultaneously in the parallel mediation framework. X = Independent Variable (SES or Personality Trait); M = Mediator; Y = Dependent Variable (Academic Performance). B = unstandardized coefficient; SE = standard error (cluster-robust for paths a, b, and c' ; bootstrap SE for indirect effects); CI = confidence interval (cluster-robust for paths a, b, and c' ; percentile bootstrap CI based on 5000 bootstrap samples for indirect effects).

Appendix C. Robustness checks

This appendix synthesizes seven comprehensive robustness checks to validate the stability and generalizability of our core findings: (1) alternative outcome measures (within-provider percentiles compared with within-provider standardized z-scores), (2) SES operationalizations (average parental income, wealth, and highest parental education), (3) AI usage measurement (CatBoost-weighted composite (AI usage intensity) compared with simple tool count), (4) personality trait operationalization (Graded Response Model compared with Exploratory Factor Analysis), (5) digital literacy operationalization (Graded Response Model compared with Exploratory Factor Analysis), (6) digital literacy components analysis (competency compared with autonomy compared with AI usage intensity), and (7) alternative analytical approach validation (PROCESS macro compared with Structural Equation Modeling). Across all specifications, the pattern of results remains stable: digital literacy consistently mediates the personality-achievement relationship, AI usage does not, and the effects of SES remain largely direct.

C.1. Robustness check 1: alternative outcome measures

We replicated the parallel mediation models replacing the main outcome (within-provider national percentiles) with within-provider standardized z-scores. All specifications, covariates, clustering, and bootstrap procedures remained identical to the main analysis. This robustness check assesses whether our conclusions depend on outcome scaling/normalization. The parallel mediation framework simultaneously tests both AI usage intensity and digital literacy as mediators for each predictor, providing a comprehensive test of our theoretical model. Results are virtually unchanged across outcome definitions. Digital literacy remains a significant mediator for personality traits, AI usage intensity remains a non-significant mediator, and direct effects are directionally and substantively similar to the main analysis.

Fig. C.1 displays the unified parallel mediation model using standardized z-scores as the outcome variable. The figure simultaneously shows all pathways from family SES and personality traits to academic performance through both AI usage intensity and digital literacy.

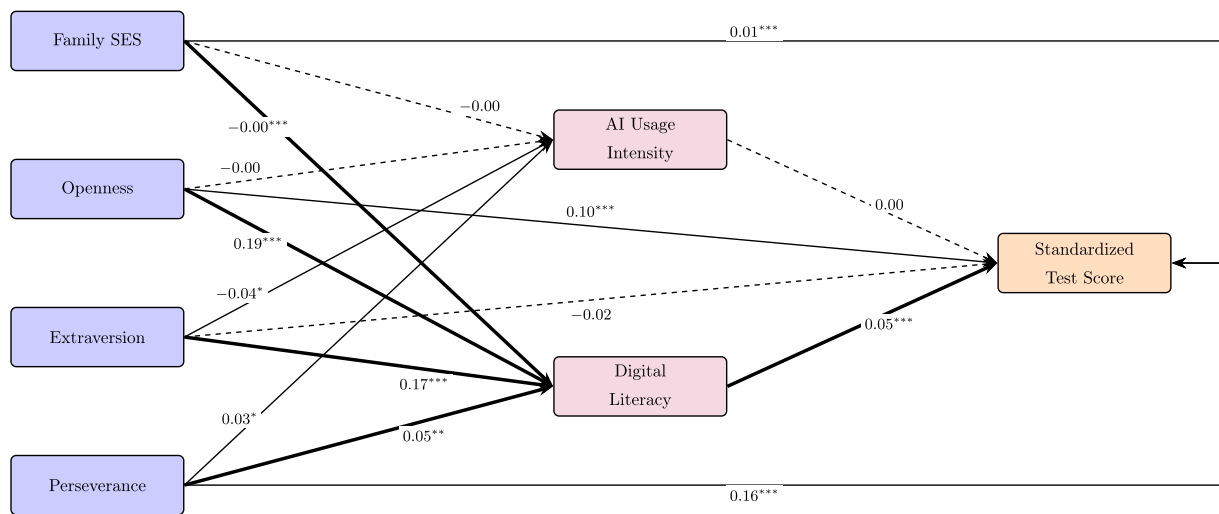


Fig. C.1. Robustness Check 1: Parallel mediation model using within-provider standardized test scores (z-scores) as the outcome ($N = 4,497$ Grade 6 students). Path coefficients are unstandardized; * $p < .05$, ** $p < .01$, *** $p < .001$. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones. Full results in Table C.1. Confidence intervals based on 5000 bootstrap samples.

Table C.1

Robustness check: Parallel mediation analyses using standardized test scores (Z-scores) as outcome variable.

Model (X → M → Y)	Path a: X → M		Path b: M → Y		Indirect Effect (a*b)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage Intensity								
SES → AI Usage → Z-Score	-.0002 (.0013)	[-.003, 0.002]	0.0037 (.0224)	[-.041, 0.048]	-.0000 (.0000)	[-.0000, 0.0000]	0.0077 (.0007)	[.006, 0.009]
Openness → AI Usage → Z-Score	-.0040 (.0222)	[-.048, 0.040]	0.0037 (.0224)	[-.041, 0.048]	-.0000 (.0002)	[-.0005, 0.0004]	0.0972 (.0149)	[.068, 0.127]
Extraversion → AI Usage → Z-Score	-.0370 (.0150)	[-.067, -.007]	0.0037 (.0224)	[-.041, 0.048]	-.0001 (.0005)	[-.001, 0.001]	-.0232 (.0150)	[-.053, 0.006]
Perseverance → AI Usage → Z-Score	0.0337 (.0146)	[.005, 0.063]	0.0037 (.0224)	[-.041, 0.048]	0.0001 (.0005)	[-.001, 0.001]	0.1584 (.0126)	[.134, 0.183]
Mediator: Digital Literacy								
SES → Digital Literacy → Z-Score	-.0049 (.0007)	[-.006, -.004]	0.0548 (.0125)	[.030, 0.080]	-.0003 (.0001)	[-.0004, -.0001]	0.0077 (.0007)	[.006, 0.009]
Openness → Digital Literacy → Z-Score	0.1865 (.0178)	[.151, 0.222]	0.0548 (.0125)	[.030, 0.080]	0.0102 (.0026)	[.005, 0.016]	0.0972 (.0149)	[.068, 0.127]
Extraversion → Digital Literacy → Z-Score	0.1679 (.0169)	[.135, 0.201]	0.0548 (.0125)	[.030, 0.080]	0.0092 (.0024)	[.005, 0.014]	-.0232 (.0150)	[-.053, 0.006]
Perseverance → Digital Literacy → Z-Score	0.0475 (.0153)	[.017, 0.078]	0.0548 (.0125)	[.030, 0.080]	0.0026 (.0011)	[.001, 0.005]	0.1584 (.0126)	[.134, 0.183]

Note. $N = 4,497$. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. Additionally, to address potential confounding, for each model we included the remaining independent variables (the other personality traits and SES) and the alternative mediator as covariates. X = Independent Variable (SES or Personality Trait); M = Mediator; Y = Dependent Variable (Academic Z-Score Performance). B = unstandardized coefficient; SE = standard error; CI = confidence interval based on 5000 bootstrap samples. Results demonstrate robustness of findings across different outcome measure operationalizations.

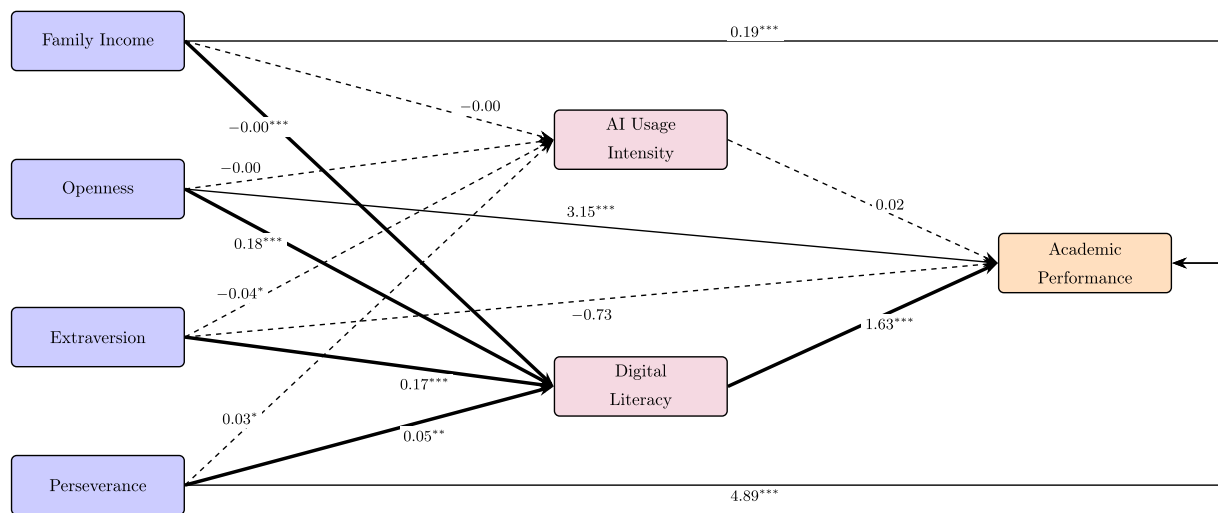


Fig. C.2. Robustness Check 2a: Parallel mediation model using average family income as the SES measure ($N = 4,497$ Grade 6 students). Path coefficients are unstandardized; * $p < .05$, ** $p < .01$, *** $p < .001$. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones. Full results in Table C.2. Confidence intervals based on 5000 bootstrap samples.

literacy mediators. The pattern closely mirrors the percentile-based results, supporting robustness to outcome operationalization. The unified diagram provides a comprehensive visualization of how both mediators operate simultaneously within the same theoretical framework.

Table C.1 reports full path estimates for the z-score specification; the results are broadly in line with those reported in Table B.1 from the main analysis, showing similar patterns of mediation and direct effects when standardized outcomes are used.

C.2. Robustness check 2: alternative SES measures

We assess sensitivity to SES measurement by re-estimating the mediation models under three different SES specifications: (1) average parental income, (2) average parental wealth, and (3) highest parental education (medium/high vs low). Model structure, covariates, clustering, and bootstrapping are identical to the main analysis. Across all SES measures, conclusions remain consistent with the main analysis. Digital literacy mediates personality–performance links; AI usage does not. SES effects on performance remain largely direct, with limited or no mediation via AI usage. Digital literacy shows statistically significant but practically negligible indirect effects for SES across all

operationalizations (meaning that higher SES is correlated with lower digital literacy, which in turn relates to lower academic performance), with effects ranging from small (income and wealth: -0.01) to more substantial for parental education (medium: -0.21 , high: -0.41). This pattern aligns with main analysis findings and suggests that SES operates through direct pathways rather than digital literacy mechanisms.

C.2.1. Visual results: family income as SES predictor

Using average parental income as the SES measure, we examined parallel mediation models to test robustness across different operationalizations of socioeconomic status. The income-based operationalization provides insights into how current family financial resources relate to digital literacy and academic achievement.

Fig. C.2 presents the unified parallel mediation model for the family income SES specification, demonstrating consistent patterns across both AI usage intensity and digital literacy mediators.

C.2.2. Visual results: family wealth as SES predictor

Using average parental wealth as the SES measure, we examined parallel mediation models to test robustness across different aspects of socioeconomic resources. The wealth-based operationalization provides

Table C.2

Alternative SES operationalization: parallel mediation results using average family income.

Model (X → M → Y)	Path a: X → M		Path b: M → Y		Indirect Effect (a*b)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage Intensity								
Openness → AI Usage → Performance	-0.0042 (0.0222)	[-0.0478, 0.0395]	0.0186 (0.6611)	[-1.2847, 1.3220]	-0.0001 (0.0063)	[-0.0144, 0.0133]	3.1471 (0.4264)	[2.3064, 3.9879]
Extraversion → AI Usage → Performance	-0.0372 (0.0152)	[-0.0671, -0.0072]	0.0186 (0.6611)	[-1.2847, 1.3220]	-0.0007 (0.0158)	[-0.0333, 0.0326]	-0.7270 (0.4521)	[-1.6183, 0.1643]
Perseverance → AI Usage → Performance	0.0337 (0.0145)	[0.0050, 0.0623]	0.0186 (0.6611)	[-1.2847, 1.3220]	0.0006 (0.0142)	[-0.0291, 0.0300]	4.8866 (0.3701)	[4.1568, 5.6163]
Family Income → AI Usage → Performance	-0.0002 (0.0011)	[-0.0024, 0.0021]	0.0186 (0.6611)	[-1.2847, 1.3220]	-0.0000 (0.0003)	[-0.0005, 0.0005]	0.1867 (0.0192)	[0.1488, 0.2246]
Mediator: Digital Literacy								
Openness → Digital Literacy → Performance	0.1838 (0.0178)	[0.1487, 0.2188]	1.6303 (0.3632)	[0.9142, 2.3464]	0.2996 (0.0772)	[0.1515, 0.4588]	3.1471 (0.4264)	[2.3064, 3.9879]
Extraversion → Digital Literacy → Performance	0.1650 (0.0169)	[0.1318, 0.1983]	1.6303 (0.3632)	[0.9142, 2.3464]	0.2690 (0.0704)	[0.1392, 0.4124]	-0.7270 (0.4521)	[-1.6183, 0.1643]
Perseverance → Digital Literacy → Performance	0.0463 (0.0154)	[0.0160, 0.0766]	1.6303 (0.3632)	[0.9142, 2.3464]	0.0755 (0.0320)	[0.0210, 0.1486]	4.8866 (0.3701)	[4.1568, 5.6163]
Family Income → Digital Literacy → Performance	-0.0038 (0.0006)	[-0.0050, -0.0026]	1.6303 (0.3632)	[0.9142, 2.3464]	-0.0062 (0.0018)	[-0.0100, -0.0030]	0.1867 (0.0192)	[0.1488, 0.2246]

Note. $N = 4,497$. Results are from parallel mediation analyses (PROCESS Model 4) where both mediators (AI Usage Intensity and Digital Literacy) are included simultaneously in each model. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. For each model, we included the remaining independent variables (the other personality traits and family income) as covariates. Path b and direct effect (c') coefficients are the same for a given independent variable across both mediator rows because both mediators are estimated simultaneously in the parallel mediation framework. X = Independent Variable (Personality Trait or Family Income); M = Mediator; Y = Dependent Variable (Academic Performance). B = unstandardized coefficient; SE = standard error (cluster-robust for paths a, b, and c'; bootstrap SE for indirect effects); CI = confidence interval (cluster-robust for paths a, b, and c'; percentile bootstrap CI based on 5000 bootstrap samples for indirect effects).

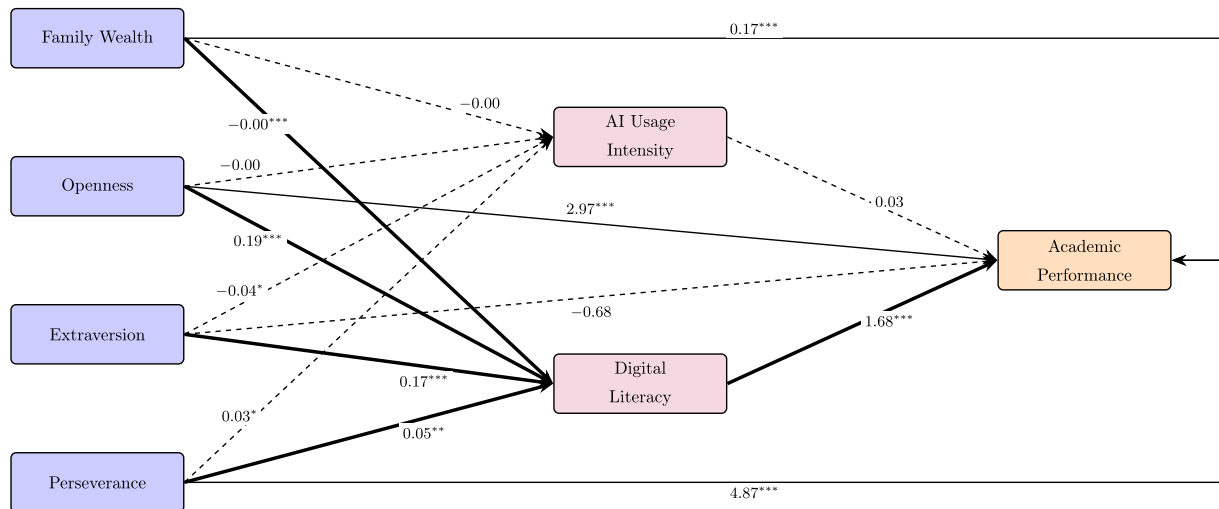


Fig. C.3. Robustness Check 2b: Parallel mediation model using average family wealth as the SES measure ($N = 4,497$ Grade 6 students). Path coefficients are unstandardized; * $p < .05$, ** $p < .01$, *** $p < .001$. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones. Full results in Table C.3. Confidence intervals based on 5000 bootstrap samples.

insights into how accumulated family assets, rather than current income, relate to digital literacy and academic achievement.

Fig. C.3 presents the unified parallel mediation model for the family wealth SES specification, demonstrating consistent patterns across both AI usage intensity and digital literacy mediators.

C.2.3. Visual results: parental education as SES predictor

Using parental education levels as dummy variables (medium and high education with low education as reference), we examined parallel mediation models to capture the cultural and human capital dimensions of SES. This specification tests whether education-based SES measures yield similar mediation patterns.

Fig. C.4 presents the unified parallel mediation model for the parental education SES specification, demonstrating consistent patterns across both AI usage intensity and digital literacy mediators.

C.3. Robustness check 3: alternative AI usage measures

To test the sensitivity of our findings to the AI usage measure, we replaced the CatBoost-weighted composite index (AI usage intensity) with a simple sum score of the total number of AI tools used. This check validates that our conclusion regarding the limited mediating role of AI usage is not an artifact of our measurement approach but holds even with a more basic operationalization. The alternative specification keeps

Table C.3

Alternative SES operationalization: parallel mediation results using average family wealth.

Model (X → M → Y)	Path a: X → M		Path b: M → Y		Indirect Effect (a*b)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage Intensity								
Openness → AI Usage → Performance	−0.0040 (0.0223)	[−0.0478, 0.0399]	0.0251 (0.6712)	[−1.2981, 1.3484]	−0.0001 (0.0063)	[−0.0143, 0.0133]	2.9718 (0.4242)	[2.1354, 3.8081]
Extraversion → AI Usage → Performance	−0.0371 (0.0151)	[−0.0668, −0.0074]	0.0251 (0.6712)	[−1.2981, 1.3484]	−0.0009 (0.0158)	[−0.0338, 0.0324]	−0.6849 (0.4443)	[−1.5607, 0.1910]
Perseverance → AI Usage → Performance	0.0337 (0.0146)	[0.0050, 0.0625]	0.0251 (0.6712)	[−1.2981, 1.3484]	0.0008 (0.0142)	[−0.0286, 0.0299]	4.8747 (0.3740)	[4.1375, 5.6120]
Family Wealth → AI Usage → Performance	−0.0002 (0.0010)	[−0.0022, 0.0019]	0.0251 (0.6712)	[−1.2981, 1.3484]	−0.0000 (0.0002)	[−0.0005, 0.0005]	0.1660 (0.0170)	[0.1324, 0.1996]
Mediator: Digital Literacy								
Openness → Digital Literacy → Performance	0.1877 (0.0178)	[0.1525, 0.2229]	1.6773 (0.3701)	[0.9476, 2.4070]	0.3149 (0.0793)	[0.1645, 0.4785]	2.9718 (0.4242)	[2.1354, 3.8081]
Extraversion → Digital Literacy → Performance	0.1653 (0.0170)	[0.1318, 0.1987]	1.6773 (0.3701)	[0.9476, 2.4070]	0.2772 (0.0709)	[0.1453, 0.4216]	−0.6849 (0.4443)	[−1.5607, 0.1910]
Perseverance → Digital Literacy → Performance	0.0470 (0.0153)	[0.0169, 0.0770]	1.6773 (0.3701)	[0.9476, 2.4070]	0.0788 (0.0328)	[0.0225, 0.1520]	4.8747 (0.3740)	[4.1375, 5.6120]
Family Wealth → Digital Literacy → Performance	−0.0039 (0.0006)	[−0.0051, −0.0026]	1.6773 (0.3701)	[0.9476, 2.4070]	−0.0065 (0.0018)	[−0.0102, −0.0033]	0.1660 (0.0170)	[0.1324, 0.1996]

Note. $N = 4,497$. Results are from parallel mediation analyses (PROCESS Model 4) where both mediators (AI Usage Intensity and Digital Literacy) are included simultaneously in each model. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. For each model, we included the remaining independent variables (the other personality traits and family wealth) as covariates. Path b and direct effect (c') coefficients are the same for a given independent variable across both mediator rows because both mediators are estimated simultaneously in the parallel mediation framework. X = Independent Variable (Personality Trait or Family Wealth); M = Mediator; Y = Dependent Variable (Academic Performance). B = unstandardized coefficient; SE = standard error (cluster-robust for paths a, b, and c'; bootstrap SE for indirect effects); CI = confidence interval (cluster-robust for paths a, b, and c'; percentile bootstrap CI based on 5000 bootstrap samples for indirect effects).

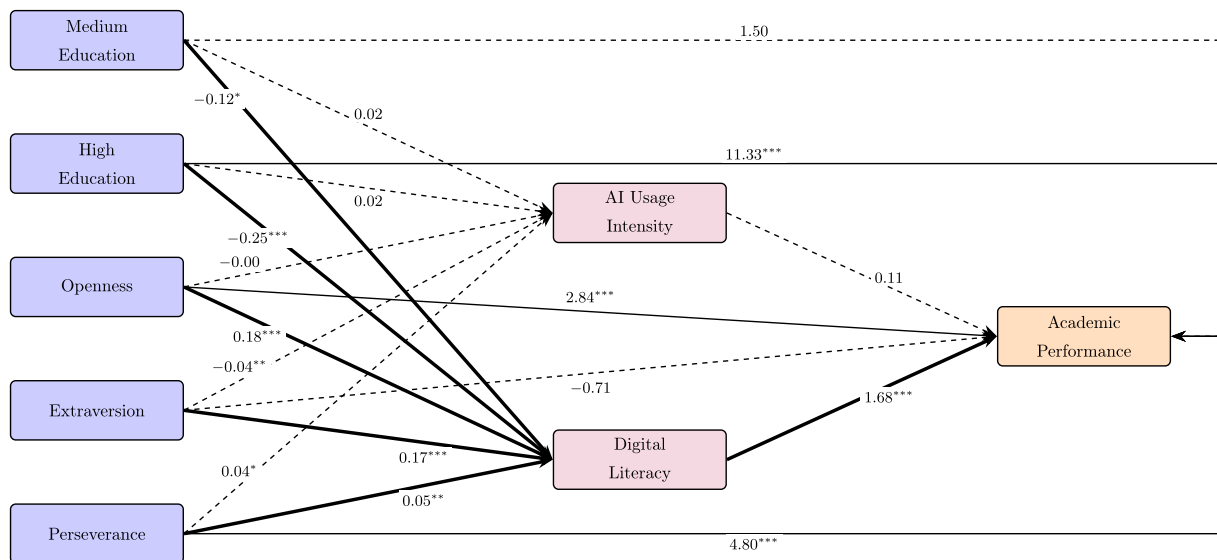


Fig. C.4. Robustness Check 2c: Parallel mediation model using parental education level as the SES measure ($N = 4,175$ Grade 6 students with complete parental education data). Path coefficients are unstandardized; * $p < .05$, ** $p < .01$, *** $p < .001$. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones. Full results in Table C.4. Confidence intervals based on 5000 bootstrap samples.

the same Hayes PROCESS Model 4 parallel mediation setup, covariates, sample clustering, and bootstrap inference as the main analysis. The only change is that the AI usage mediator is a simple sum score of tools used rather than the CatBoost-weighted composite (AI usage intensity). All models retain identical control variables (oldest child status, IQ, gender, age, traditional ICT usage, and test provider indicators) and use cluster-robust standard errors with 5000 bootstrap samples for confidence interval estimation.

The simple AI usage count measure produced virtually identical patterns to the CatBoost-weighted measure. AI usage remained a non-significant mediator across all four predictors, with confidence

intervals consistently including zero. Conversely, digital literacy remained a robust and significant mediator for all personality traits (openness, extraversion, perseverance) and SES, with confidence intervals excluding zero. This robustness check provides strong evidence that our substantive conclusions about differential mediation mechanisms are not dependent on the sophistication of AI usage measurement but reflect genuine differences in how AI usage patterns versus digital competencies translate into academic benefits.

Fig. C.5 presents the unified parallel mediation model for the alternative AI usage specification (simple sum score), demonstrating the consistent patterns across both AI usage and digital literacy mediators.

Table C.4

Alternative SES operationalization: parallel mediation results using parental education levels.

Model (X → M → Y)	Path a: X → M		Path b: M → Y		Indirect Effect (a*b)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage Intensity								
Openness → AI Usage → Performance	−0.0021 (0.0223)	[−0.0460, 0.0418]	0.1107 (0.6646)	[−1.1996, 1.4209]	−0.0002 (0.0069)	[−0.0159, 0.0144]	2.8432 (0.4285)	[1.9984, 3.6880]
Extraversion → AI Usage → Performance	−0.0415 (0.0154)	[−0.0720, −0.0111]	0.1107 (0.6646)	[−1.1996, 1.4209]	−0.0046 (0.0180)	[−0.0443, 0.0302]	−0.7146 (0.4534)	[−1.6085, 0.1793]
Perseverance → AI Usage → Performance	0.0366 (0.0156)	[0.0058, 0.0674]	0.1107 (0.6646)	[−1.1996, 1.4209]	0.0041 (0.0156)	[−0.0268, 0.0394]	4.8020 (0.3727)	[4.0673, 5.5368]
Medium Education → AI Usage → Performance	0.0178 (0.0594)	[−0.0993, 0.1349]	0.1107 (0.6646)	[−1.1996, 1.4209]	0.0020 (0.0229)	[−0.0461, 0.0548]	1.4995 (1.3801)	[−1.2213, 4.2202]
High Education → AI Usage → Performance	0.0203 (0.0774)	[−0.1324, 0.1729]	0.1107 (0.6646)	[−1.1996, 1.4209]	0.0022 (0.0227)	[−0.0452, 0.0559]	11.3328 (1.3443)	[8.6826, 13.9830]
Mediator: Digital Literacy								
Openness → Digital Literacy → Performance	0.1784 (0.0170)	[0.1448, 0.2119]	1.6753 (0.3901)	[0.9062, 2.4444]	0.2988 (0.0767)	[0.1529, 0.4552]	2.8432 (0.4285)	[1.9984, 3.6880]
Extraversion → Digital Literacy → Performance	0.1686 (0.0174)	[0.1342, 0.2029]	1.6753 (0.3901)	[0.9062, 2.4444]	0.2824 (0.0737)	[0.1441, 0.4300]	−0.7146 (0.4534)	[−1.6085, 0.1793]
Perseverance → Digital Literacy → Performance	0.0512 (0.0163)	[0.0191, 0.0833]	1.6753 (0.3901)	[0.9062, 2.4444]	0.0858 (0.0345)	[0.0255, 0.1601]	4.8020 (0.3727)	[4.0673, 5.5368]
Medium Education → Digital Literacy → Performance	−0.1228 (0.0498)	[−0.2210, −0.0247]	1.6753 (0.3901)	[0.9062, 2.4444]	−0.2058 (0.1020)	[−0.4270, −0.0278]	1.4995 (1.3801)	[−1.2213, 4.2202]
High Education → Digital Literacy → Performance	−0.2471 (0.0522)	[−0.3500, −0.1443]	1.6753 (0.3901)	[0.9062, 2.4444]	−0.4140 (0.1316)	[−0.6958, −0.1809]	11.3328 (1.3443)	[8.6826, 13.9830]

Note. $N = 4,175$. Results are from parallel mediation analyses (PROCESS Model 4) where both mediators (AI Usage Intensity and Digital Literacy) are included simultaneously in each model. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. For each model, we included the remaining independent variables (the other personality traits and education levels) as covariates. Path b and direct effect (c') coefficients are the same for a given independent variable across both mediator rows because both mediators are estimated simultaneously in the parallel mediation framework. Education levels are compared to low education reference group. X = Independent Variable (Personality Trait or Parental Education); M = Mediator; Y = Dependent Variable (Academic Performance). B = unstandardized coefficient; SE = standard error (cluster-robust for paths a, b, and c'; bootstrap SE for indirect effects); CI = confidence interval (cluster-robust for paths a, b, and c'; percentile bootstrap CI based on 5000 bootstrap samples for indirect effects).

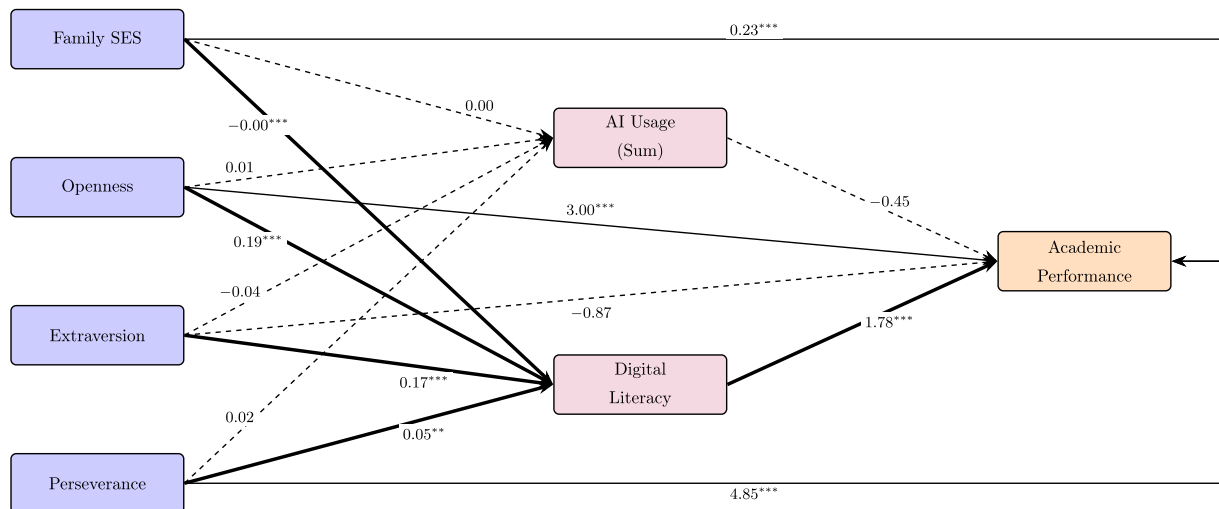


Fig. C.5. Robustness Check 3: Parallel mediation model using a simple sum score of AI tool usage in place of the CatBoost-weighted index ($N = 4,497$ Grade 6 students). Path coefficients are unstandardized; * $p < .05$, ** $p < .01$, *** $p < .001$. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones. Full results in Table C.5. Confidence intervals based on 5000 bootstrap samples.

Table C.5 provides the detailed numerical results across all eight models examined, confirming the visual patterns shown in the path diagrams above.

C.4. Robustness check 4: EFA-derived personality traits

We replicated the mediation analyses using personality traits derived from Exploratory Factor Analysis (EFA) instead of the Graded Response Model (GRM) used in the main analyses. This check assessed

whether our findings depend on the specific psychometric modeling approach employed for personality trait construction. The EFA-derived personality traits were constructed using the same questionnaire items as the main analysis but processed through principal axis factoring (a communality-based extraction method that extracts shared variance only) to derive a single unidimensional trait factor for each personality scale. Factor scores were computed using the regression method, which produces standardized trait estimates comparable to the GRM-based composite scores used in the main analysis. This classical factor

Table C.5

Alternative AI usage operationalization: comprehensive robustness results.

Model ($X \rightarrow M \rightarrow Y$)	Path a: $X \rightarrow M$		Path b: $M \rightarrow Y$		Indirect Effect ($a*b$)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage (Sum Score)								
Openness $\rightarrow$ AI Usage $\rightarrow$ Performance	0.0101 (0.0275)	[-0.0441, 0.0642]	-0.4546 (0.5091)	[-1.4582, 0.5489]	-0.0046 (0.0115)	[-0.0324, 0.0152]	3.0012 (0.4263)	[2.1606, 3.8417]
Extraversion $\rightarrow$ AI Usage $\rightarrow$ Performance	-0.0394 (0.0226)	[-0.0840, 0.0053]	-0.4546 (0.5091)	[-1.4582, 0.5489]	0.0179 (0.0169)	[-0.0060, 0.0586]	-0.8681 (0.4462)	[-1.7478, 0.0115]
Perseverance $\rightarrow$ AI Usage $\rightarrow$ Performance	0.0241 (0.0195)	[-0.0143, 0.0626]	-0.4546 (0.5091)	[-1.4582, 0.5489]	-0.0110 (0.0121)	[-0.0410, 0.0074]	4.8458 (0.3718)	[4.1128, 5.5789]
SES $\rightarrow$ AI Usage $\rightarrow$ Performance	0.0006 (0.0017)	[-0.0028, 0.0039]	-0.4546 (0.5091)	[-1.4582, 0.5489]	-0.0003 (0.0005)	[-0.0014, 0.0006]	0.2256 (0.0211)	[0.1840, 0.2671]
Mediator: Digital Literacy								
Openness $\rightarrow$ Digital Literacy $\rightarrow$ Performance	0.1865 (0.0178)	[0.1514, 0.2215]	1.7828 (0.3701)	[1.0532, 2.5124]	0.3324 (0.0792)	[0.1800, 0.4957]	3.0012 (0.4263)	[2.1606, 3.8417]
Extraversion $\rightarrow$ Digital Literacy $\rightarrow$ Performance	0.1679 (0.0169)	[0.1345, 0.2012]	1.7828 (0.3701)	[1.0532, 2.5124]	0.2993 (0.0725)	[0.1653, 0.4473]	-0.8681 (0.4462)	[-1.7478, 0.0115]
Perseverance $\rightarrow$ Digital Literacy $\rightarrow$ Performance	0.0475 (0.0153)	[0.0174, 0.0777]	1.7828 (0.3701)	[1.0532, 2.5124]	0.0847 (0.0341)	[0.0257, 0.1618]	4.8458 (0.3718)	[4.1128, 5.5789]
SES $\rightarrow$ Digital Literacy $\rightarrow$ Performance	-0.0049 (0.0007)	[-0.0063, -0.0035]	1.7828 (0.3701)	[1.0532, 2.5124]	-0.0088 (0.0023)	[-0.0134, -0.0047]	0.2256 (0.0211)	[0.1840, 0.2671]

Note. $N = 4,497$. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. X = Independent Variable (Personality Trait or SES); M = Mediator; Y = Dependent Variable (Academic Performance). B = unstandardized coefficient; SE = standard error; CI = confidence interval based on 5000 bootstrap samples. Results demonstrate robustness of main findings to alternative AI usage operationalization.

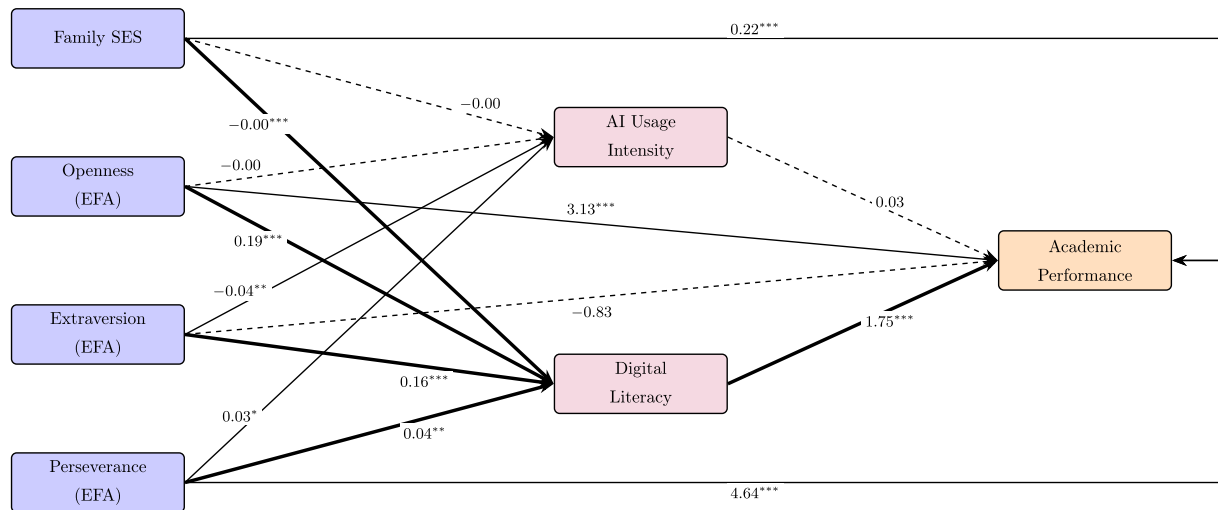


Fig. C.6. Robustness Check 4: Parallel Mediation Model Using EFA-derived Personality Traits. Path diagram shows the parallel mediation framework testing both AI Usage Intensity and Digital Literacy simultaneously as mediators across four models (Openness-EFA, Extraversion-EFA, Perseverance-EFA, and Family SES). Digital literacy shows significant indirect effects across all four models, while AI usage shows non-significant indirect effects for all predictors. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones; solid curved arrows show direct effects (c'). Full results in Table C.6. All models control for student demographics, cognitive ability, and test provider effects ($N = 4,497$ students, 207 schools).

analytic approach directly contrasts with the IRT-based GRM method that models item-level response probabilities through latent trait theory. Polychoric correlations (appropriate for ordinal Likert-scale data) were used as input to the factor model to account for the discrete nature of the survey items. This alternative specification maintains identical model structures, control variables, clustering procedures, and bootstrap methods, isolating the effect of measurement operationalization while systematically comparing mediation effects across all predictor-mediator combinations.

Results demonstrate consistency with the main findings regardless of personality trait operationalization. EFA-derived personality traits show virtually identical patterns of mediation effects, with digital literacy consistently mediating the personality-achievement relationship while AI usage intensity shows non-significant mediation across all models. The robustness of these patterns across different psychometric

approaches strengthens confidence in the substantive conclusions about personality-driven digital divides.

Fig. C.6 presents the parallel mediation path diagram for the EFA-derived personality traits specification, demonstrating consistent patterns across both AI usage and digital literacy mediators.

Table C.6 provides the detailed numerical results across all eight models examined, confirming the visual patterns shown in the path diagrams above.

C.5. Robustness check 5: using EFA-derived digital literacy

We replicated the mediation analyses using digital literacy measures derived from Exploratory Factor Analysis (EFA) instead of the Graded Response Model (GRM) used in the main analyses. This check assessed whether our findings depend on the

Table C.6
Robustness check 4: parallel mediation results using EFA-derived personality traits.

Model (X → M → Y)	Path a: X → M		Path b: M → Y		Indirect Effect (a*b)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage Intensity								
Openness (EFA) → AI Usage → Performance	-0.0033 (0.0212)	[-0.0452, 0.0386]	0.0302 (0.6605)	[-1.2720, 1.3324]	-0.0001 (0.0060)	[-0.0137, 0.0125]	3.1251 (0.4057)	[2.3252, 3.9250]
Extraversion (EFA) → AI Usage → Performance	-0.0371 (0.0142)	[-0.0652, -0.0091]	0.0302 (0.6605)	[-1.2720, 1.3324]	-0.0011 (0.0156)	[-0.0336, 0.0311]	-0.8308 (0.4224)	[-1.6636, 0.0020]
Perseverance (EFA) → AI Usage → Performance	0.0324 (0.0142)	[0.0044, 0.0604]	0.0302 (0.6605)	[-1.2720, 1.3324]	0.0010 (0.0135)	[-0.0269, 0.0286]	4.6371 (0.3600)	[3.9273, 5.3469]
Family SES → AI Usage → Performance	-0.0002 (0.0013)	[-0.0028, 0.0024]	0.0302 (0.6605)	[-1.2720, 1.3324]	-0.0000 (0.0003)	[-0.0006, 0.0006]	0.2238 (0.0210)	[0.1823, 0.2653]
Mediator: Digital Literacy								
Openness (EFA) → Digital Literacy → Performance	0.1851 (0.0170)	[0.1515, 0.2187]	1.7477 (0.3689)	[1.0203, 2.4751]	0.3234 (0.0780)	[0.1745, 0.4854]	3.1251 (0.4057)	[2.3252, 3.9250]
Extraversion (EFA) → Digital Literacy → Performance	0.1584 (0.0163)	[0.1262, 0.1906]	1.7477 (0.3689)	[1.0203, 2.4751]	0.2768 (0.0682)	[0.1498, 0.4164]	-0.8308 (0.4224)	[-1.6636, 0.0020]
Perseverance (EFA) → Digital Literacy → Performance	0.0393 (0.0147)	[0.0103, 0.0683]	1.7477 (0.3689)	[1.0203, 2.4751]	0.0687 (0.0311)	[0.0141, 0.1385]	4.6371 (0.3600)	[3.9273, 5.3469]
Family SES → Digital Literacy → Performance	-0.0049 (0.0007)	[-0.0063, -0.0035]	1.7477 (0.3689)	[1.0203, 2.4751]	-0.0086 (0.0022)	[-0.0132, -0.0045]	0.2238 (0.0210)	[0.1823, 0.2653]

Note. $N = 4,497$. Results are from parallel mediation analyses (PROCESS Model 4) where both mediators (AI Usage Intensity and Digital Literacy) were tested simultaneously in each model. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. Personality traits were derived using Exploratory Factor Analysis (EFA) as a robustness check against the main analysis using Item Response Theory (IRT-based Graded Response Model). Path b and direct effect c' coefficients are identical for a given independent variable across both mediator rows because both mediators were modeled simultaneously. Bootstrap confidence intervals (5,000 samples with replacement) were used for indirect effects. Standard errors were clustered at the school level (207 schools).

specific psychometric modeling approach employed for digital literacy measurement.

The EFA-derived digital literacy measure was constructed using the same questionnaire items as the main analysis but processed through principal axis factoring (a communality-based extraction method that extracts shared variance only) to derive a single unidimensional digital literacy factor. Factor scores were computed using the regression method, which produces standardized estimates comparable to the GRM-based composite scores used in the main analysis. This classical factor analytic approach directly contrasts with the IRT-based GRM method that models item-level response probabilities through latent trait theory. Polychoric correlations (appropriate for ordinal Likert-scale data) were used as input to the factor model to account for the discrete nature of the survey items. This alternative specification maintains identical model structures, control variables, clustering procedures, and bootstrap methods, isolating the effect of measurement operationalization while systematically comparing mediation effects across all predictor-mediator combinations.

Results demonstrate consistency with the main findings regardless of digital literacy operationalization. EFA-derived digital literacy shows virtually identical patterns of mediation effects, with digital literacy consistently mediating the personality-achievement relationship while AI usage intensity shows non-significant mediation across all models. The robustness of these patterns across different psychometric approaches strengthens confidence in the substantive conclusions about personality-driven digital divides.

Fig. C.7 presents the parallel mediation path diagram for the EFA-derived digital literacy specification, demonstrating consistent patterns across both AI usage and digital literacy mediators.

Table C.7 provides the detailed numerical results across all eight models examined, confirming the visual patterns shown in the path diagrams above.

C.6. Robustness check 6: digital literacy components analysis

To examine the sensitivity of our digital literacy mediation findings to measurement specification, we conducted alternative analyses by decomposing the composite digital literacy measure into its two constituent dimensions: digital competency and digital autonomy.

These parallel mediation models test whether the overall mediation effect of digital literacy operates uniformly across its technical proficiency component (digital competency) and its self-regulation and autonomy component (digital autonomy), or whether different aspects of digital literacy account for the observed mediation effects.

The alternative PROCESS macro specifications employed the same Model 4 parallel mediation framework and covariates as the main analyses. However, they replaced the composite digital literacy measure with its two constituent dimensions (digital competency and digital autonomy) alongside AI usage intensity, yielding three separate mediators tested simultaneously. This approach examines the robustness of our findings against measurement choices and provides insight into which specific mechanisms account for the observed mediation effects.

Results demonstrate that digital competency and digital autonomy both serve as significant mediators, while AI usage intensity again shows minimal mediation effects. As shown in Fig. C.8 and Table C.8, digital competency shows significant indirect effects for openness (indirect effect = 0.201, 95% CI [0.043, 0.364]), extraversion (indirect effect = 0.198, 95% CI [0.043, 0.364]), and perseverance (indirect effect = 0.056, 95% CI [0.010, 0.121]). Digital autonomy also shows significant indirect effects for openness (indirect effect = 0.124, 95% CI [0.013, 0.247]) and extraversion (indirect effect = 0.084, 95% CI [0.009, 0.176]). Family SES shows significant negative indirect effects through both digital competency and digital autonomy. These findings suggest that technical digital skills and autonomous digital usage, rather than machine learning-derived usage patterns, primarily account for the observed mediation effects.

C.7. Robustness check 7: structural equation modeling (SEM) validation

To assess the robustness of our findings to alternative analytical approaches, we replicated key mediation pathways using Structural Equation Modeling (SEM). While the main analyses employed the PROCESS macro for theory-driven path analysis, SEM provides complementary validation through simultaneous estimation of all pathways and comprehensive model fit assessment.

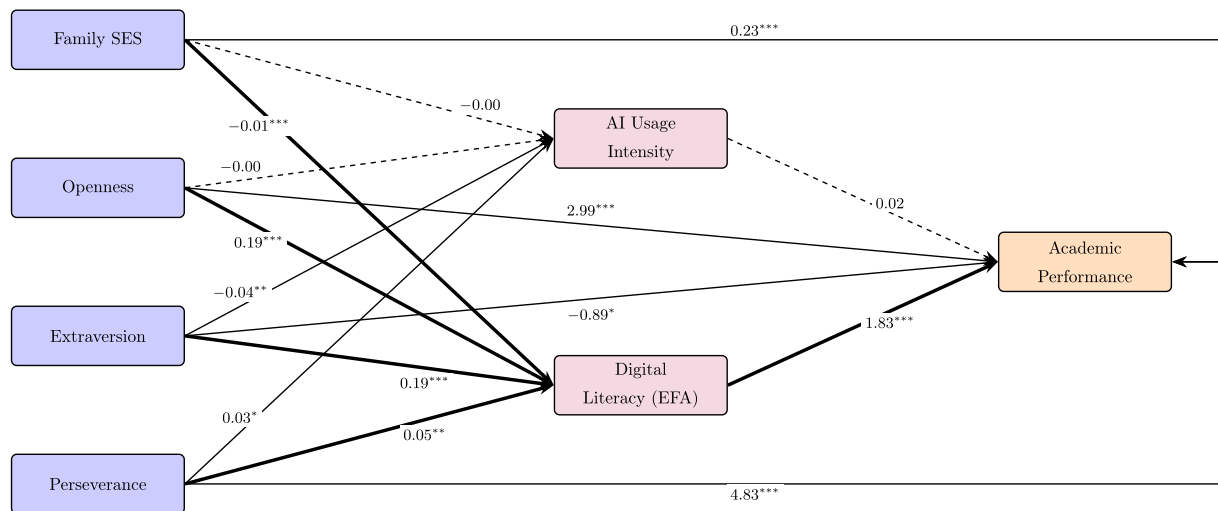


Fig. C.7. Robustness Check 5: Parallel Mediation Model Using EFA-derived Digital Literacy. Path diagram shows the parallel mediation framework testing both AI Usage Intensity and Digital Literacy (EFA-derived) simultaneously as mediators across four models (Openness, Extraversion, Perseverance, and Family SES). Digital literacy (EFA) shows significant indirect effects across all four models, while AI usage shows non-significant indirect effects for all predictors. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones; solid curved arrows show direct effects (c'). Full results in Table C.7. All models control for student demographics, cognitive ability, and test provider effects ($N = 4,497$ students, 207 schools).

Table C.7
Robustness check 5: parallel mediation results using EFA-derived digital literacy.

Model ($X \rightarrow M \rightarrow Y$)	Path a: $X \rightarrow M$		Path b: $M \rightarrow Y$		Indirect Effect ($a*b$)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI	B (SE)	95% CI
Mediator: AI Usage Intensity								
Openness $\rightarrow$ AI Usage $\rightarrow$ Performance	-0.0040 (0.0222)	[-0.0479, 0.0398]	0.0242 (0.6610)	[-1.2789, 1.3273]	-0.0001 (0.0063)	[-0.0144, 0.0129]	2.9855 (0.4256)	[2.1464, 3.8246]
Extraversion $\rightarrow$ AI Usage $\rightarrow$ Performance	-0.0370 (0.0150)	[-0.0666, -0.0074]	0.0242 (0.6610)	[-1.2789, 1.3273]	-0.0009 (0.0157)	[-0.0337, 0.0316]	-0.8892 (0.4473)	[-1.7711, -0.0073]
Perseverance $\rightarrow$ AI Usage $\rightarrow$ Performance	0.0337 (0.0146)	[0.0050, 0.0625]	0.0242 (0.6610)	[-1.2789, 1.3273]	0.0008 (0.0141)	[-0.0286, 0.0298]	4.8260 (0.3730)	[4.0905, 5.5614]
Family SES $\rightarrow$ AI Usage $\rightarrow$ Performance	-0.0002 (0.0013)	[-0.0028, 0.0024]	0.0242 (0.6610)	[-1.2789, 1.3273]	-0.0000 (0.0003)	[-0.0006, 0.0006]	0.2264 (0.0211)	[0.1847, 0.2680]
Mediator: Digital Literacy (EFA-derived)								
Openness $\rightarrow$ Digital Literacy (EFA) $\rightarrow$ Performance	0.1876 (0.0189)	[0.1503, 0.2249]	1.8316 (0.3479)	[1.1456, 2.5175]	0.3436 (0.0763)	[0.1975, 0.5013]	2.9855 (0.4256)	[2.1464, 3.8246]
Extraversion $\rightarrow$ Digital Literacy (EFA) $\rightarrow$ Performance	0.1852 (0.0178)	[0.1500, 0.2203]	1.8316 (0.3479)	[1.1456, 2.5175]	0.3391 (0.0759)	[0.1995, 0.4943]	-0.8892 (0.4473)	[-1.7711, -0.0073]
Perseverance $\rightarrow$ Digital Literacy (EFA) $\rightarrow$ Performance	0.0507 (0.0159)	[0.0193, 0.0821]	1.8316 (0.3479)	[1.1456, 2.5175]	0.0928 (0.0362)	[0.0287, 0.1746]	4.8260 (0.3730)	[4.0905, 5.5614]
Family SES $\rightarrow$ Digital Literacy (EFA) $\rightarrow$ Performance	-0.0053 (0.0008)	[-0.0068, -0.0038]	1.8316 (0.3479)	[1.1456, 2.5175]	-0.0098 (0.0023)	[-0.0146, -0.0056]	0.2264 (0.0211)	[0.1847, 0.2680]

Note. $N = 4,497$. Results are from parallel mediation analyses (PROCESS Model 4) where both mediators (AI Usage Intensity and Digital Literacy-EFA) were tested simultaneously in each model. All models controlled for student sex, age, IQ, traditional ICT-Ed Usage, oldest child status, and test provider. Digital literacy was derived using Exploratory Factor Analysis (EFA) as a robustness check against the main analysis using Item Response Theory (IRT-based Graded Response Model). Path b and direct effect c' coefficients are identical for a given independent variable across both mediator rows because both mediators were modeled simultaneously. Bootstrap confidence intervals (5,000 samples with replacement) were used for indirect effects. Standard errors were clustered at the school level (207 schools).

C.7.1. Structural equation modeling versus PROCESS: methodological comparison

The main analyses used the PROCESS macro (Model 4) rather than SEM for two primary methodological reasons. First, our study focuses on testing specific, theory-driven mediation pathways rather than assessing the global fit of a complex theoretical model. The PROCESS macro is explicitly designed for this path-by-path approach, allowing precise examination of individual indirect effects with bootstrap confidence intervals (Hayes, 2022). Second, our key variables are operationalized as observed composite scores derived from psychometric scaling (IRT and CatBoost), not latent constructs requiring measurement model specification, making the more parsimonious PROCESS approach both suitable and statistically powerful for our clearly defined single-mediator models.

However, SEM offers several complementary advantages that provide valuable robustness validation. SEM enables simultaneous estimation of all pathways, reducing concerns about multiple testing and providing global model fit assessment through established indices (CFI, TLI, RMSEA, SRMR). Additionally, SEM can explicitly model measurement error and test structural invariance across different specifications, offering a more comprehensive evaluation of model adequacy.

In our case, SEM serves as a robustness check rather than the primary analytical approach because: (1) it confirms that conclusions hold under simultaneous pathway estimation, (2) it demonstrates adequate model fit despite the complexity of modeling all relationships simultaneously, and (3) it validates that the theoretical framework adequately represents the observed covariance structure among variables.

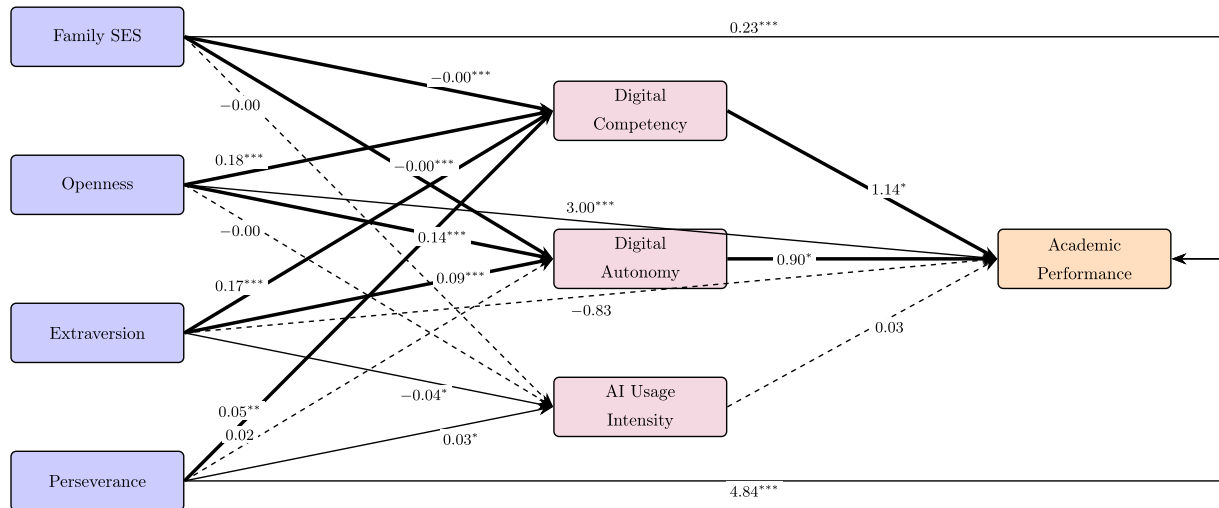


Fig. C.8. Robustness Check 6: Parallel Mediation Model Using Digital Competency, Digital Autonomy, and AI Usage Intensity. Path diagram shows the parallel mediation framework testing three digital literacy-related mediators simultaneously: digital competency (technical proficiency), digital autonomy (self-regulation), and AI usage intensity (machine learning-derived usage patterns). Across all four models (Openness, Extraversion, Perseverance, and Family SES), digital competency and digital autonomy show significant indirect effects, while AI usage intensity shows non-significant mediation. Bold paths indicate significant indirect effects; dashed paths indicate non-significant ones; solid curved arrows show direct effects (c'). Full results in Table C.8. All models control for student demographics, cognitive ability, and test provider effects ($N = 4,497$ students, 207 schools).

Table C.8

Robustness check 6: parallel mediation results using digital literacy components (Digital Competency, Digital Autonomy, AI Usage Intensity).

Model ($X \rightarrow M \rightarrow Y$)	Path a: $X \rightarrow M$		Path b: $M \rightarrow Y$		Indirect Effect ($a \times b$)		Direct Effect (c')	
	B (SE)	95% CI	B (SE)	95% CI	Effect	95% CI	B (SE)	95% CI
Digital Competency as Mediator								
Openness $\rightarrow$ Digital Competency $\rightarrow$ Performance	0.177 (.018)	[.142, 0.213]	1.135 (.455)	[.238, 2.032]	0.201	[.043, 0.364]	3.004 (.425)	[2.166, 3.841]
Extraversion $\rightarrow$ Digital Competency $\rightarrow$ Performance	0.174 (.018)	[.140, 0.209]	1.135 (.455)	[.238, 2.032]	0.198	[.043, 0.364]	-.832 (.453)	[-1.725, 0.062]
Perseverance $\rightarrow$ Digital Competency $\rightarrow$ Performance	0.049 (.015)	[.020, 0.079]	1.135 (.455)	[.238, 2.032]	0.056	[.010, 0.121]	4.841 (.372)	[4.108, 5.573]
Family SES $\rightarrow$ Digital Competency $\rightarrow$ Performance	-.004 (.001)	[-.005, -.002]	1.135 (.455)	[.238, 2.032]	-.004	[-.008, -.001]	0.225 (.021)	[.183, 0.267]
Digital Autonomy as Mediator								
Openness $\rightarrow$ Digital Autonomy $\rightarrow$ Performance	0.138 (.017)	[.105, 0.170]	0.903 (.445)	[.026, 1.781]	0.124	[.013, 0.247]	3.004 (.425)	[2.166, 3.841]
Extraversion $\rightarrow$ Digital Autonomy $\rightarrow$ Performance	0.093 (.019)	[.056, 0.131]	0.903 (.445)	[.026, 1.781]	0.084	[.009, 0.176]	-.832 (.453)	[-1.725, 0.062]
Perseverance $\rightarrow$ Digital Autonomy $\rightarrow$ Performance	0.024 (.015)	[-.006, 0.054]	0.903 (.445)	[.026, 1.781]	0.022	[-.007, 0.069]	4.841 (.372)	[4.108, 5.573]
Family SES $\rightarrow$ Digital Autonomy $\rightarrow$ Performance	-.005 (.001)	[-.006, -.003]	0.903 (.445)	[.026, 1.781]	-.004	[-.008, -.0004]	0.225 (.021)	[.183, 0.267]
AI Usage Intensity as Mediator								
Openness $\rightarrow$ AI Usage Intensity $\rightarrow$ Performance	-.004 (.022)	[-.048, 0.040]	0.032 (.662)	[-1.273, 1.338]	-.0001	[-.014, 0.013]	3.004 (.425)	[2.166, 3.841]
Extraversion $\rightarrow$ AI Usage Intensity $\rightarrow$ Performance	-.037 (.015)	[-.067, -.007]	0.032 (.662)	[-1.273, 1.338]	-.001	[-.035, 0.031]	-.832 (.453)	[-1.725, 0.062]
Perseverance $\rightarrow$ AI Usage Intensity $\rightarrow$ Performance	0.034 (.015)	[.005, 0.063]	0.032 (.662)	[-1.273, 1.338]	0.001	[-.028, 0.030]	4.841 (.372)	[4.108, 5.573]
Family SES $\rightarrow$ AI Usage Intensity $\rightarrow$ Performance	-.000 (.001)	[-.003, 0.002]	0.032 (.662)	[-1.273, 1.338]	-.0000	[-.0006, 0.0006]	0.225 (.021)	[.183, 0.267]

Note. $N = 4,497$. All models controlled for student sex, age, IQ, traditional ICT-Ed usage, oldest child status, test provider, and the other mediators in the parallel model. B = unstandardized coefficient; SE = cluster-robust standard error; CI = confidence interval based on 5000 bootstrap samples. Digital Competency and Digital Autonomy are subscales derived from the digital literacy measure using IRT, while AI Usage Intensity represents machine learning-derived AI usage intensity from gradient boosting model predictions. Significant mediation is indicated by confidence intervals excluding zero.

C.7.2. SEM model specification and fit

We specified a structural equation model that parallels the PROCESS macro analyses, incorporating the same predictor variables (openness, extraversion, perseverance, family SES), mediators (AI usage intensity, digital literacy), outcome variable (academic performance), and control variables. The measurement model treated personality traits and digital literacy as latent constructs with their constituent observed indicators, while maintaining AI usage intensity and academic performance as observed variables consistent with their operational definitions.

Model fit was evaluated using multiple established indices. As shown in Table C.9, the model achieved acceptable fit on most criteria: RMSEA = 0.077 (90% CI [0.074, 0.080]) falls within the acceptable range (≤ 0.08), and SRMR = 0.060 indicates good fit (≤ 0.08). However, CFI = 0.840 and TLI = 0.779 fall slightly below conventional thresholds

(≥ 0.90), reflecting the complexity of simultaneously modeling multiple theoretical constructs and their interrelationships. The significant χ^2 statistic (1,975.112, $p < 0.001$) is expected given the large sample size ($N = 4,497$), where even trivial model misspecifications become statistically detectable.

C.7.3. Structural path results and validation

The SEM structural path coefficients, presented in Table C.10, demonstrate consistency with the PROCESS macro results. Digital literacy emerges as a robust predictor of academic performance ($B = 1.737$, $SE = 0.480$, $p < 0.001$), while AI usage intensity shows no significant relationship with academic outcomes ($B = 0.100$, $SE = 0.355$, $p = 0.777$), confirming our primary finding about the differential importance of digital competencies versus mere tool usage.

Table C.9

SEM robustness check: model fit indices.

Fit Measure	Value
χ^2	1,975.112
df	76.000
p-value	0.000
CFI	0.840
TLI	0.779
RMSEA	0.077
RMSEA 90% CI	[0.074, 0.080]
SRMR	0.060

Note: CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Square Residual.

Personality trait effects on digital literacy are consistently significant and substantial: openness ($B = 0.168$, $SE = 0.017$, $p < 0.001$), extraversion ($B = 0.191$, $SE = 0.018$, $p < 0.001$), and perseverance ($B = 0.060$, $SE = 0.016$, $p < 0.001$). The direct effects of personality traits on academic performance remain significant for openness ($B = 2.889$, $SE = 0.421$, $p < 0.001$) and perseverance ($B = 4.825$, $SE = 0.387$, $p < 0.001$), while extraversion shows no significant direct effect ($B = -0.619$, $SE = 0.440$, $p = 0.160$), indicating partial rather than complete mediation for openness and perseverance, consistent with PROCESS findings.

Importantly, the SEM results confirm that family SES maintains a strong direct relationship with academic performance ($B = 0.442$, $SE = 0.034$, $p < 0.001$) while showing an unexpected negative association with digital literacy ($B = -0.007$, $SE = 0.001$, $p < 0.001$), replicating the pattern observed in the main analyses and supporting our conclusion about the limited role of digital factors in mediating socioeconomic achievement gaps.

C.7.4. SEM path diagram

Fig. C.9 presents the complete SEM path diagram, displaying standardized coefficients and significance levels for all hypothesized relationships. The visual representation confirms the theoretical structure underlying our mediation analyses while highlighting the differential patterns of association between predictors, mediators, and academic outcomes.

C.8. Synthesis of robustness checks and implications

The mediation patterns remain stable regardless of how AI usage intensity and digital literacy are measured. AI usage intensity remains a null mediator, while digital literacy remains a positive mediator for personality pathways and a statistically significant but practically negligible negative mediator for SES pathways. Replacing the CatBoost-weighted AI usage index with a simple sum score produces virtually identical patterns of non-significant mediation, confirming that the limited mediating role of AI usage reflects genuine differences in how AI engagement translates into academic benefits rather than an artifact of measurement complexity. Likewise, across income, wealth, and education-based operationalizations of SES, digital literacy consistently emerges as a significant mediator for personality-academic performance relationships while AI usage shows minimal mediation effects. These patterns persist whether SES is conceptualized through economic resources (income, wealth) or cultural capital (education).

GRM-derived and EFA-derived versions of both personality traits and digital literacy yield virtually identical mediation patterns. The EFA-derived personality traits replicate the finding that digital literacy mediates the personality-achievement relationship across all three traits (openness, extraversion, perseverance), while AI usage intensity shows non-significant mediation in both specifications. The EFA-derived digital literacy measure similarly produces consistent mediation effects across all four predictors, with bootstrap confidence intervals that exclude zero. This convergence across different psychometric modeling approaches provides strong evidence that the mediating role of digital competencies is robust to methodological choices in scale construction.

Table C.10

SEM robustness check: structural path coefficients.

Dependent variable	Independent variable	Raw Est.	Std. Error	z-value	p-value	β (Stand.)	95% CI
<i>Predicting AI Usage</i>							
AI Usage	Openness	0.010	0.016	0.625	0.532	0.009	[-0.022, 0.042]
AI Usage	Extraversion	-0.052	0.017	-3.037	0.002	-0.045	[-0.086, -0.019]**
AI Usage	Perseverance	0.039	0.015	2.683	0.007	0.038	[0.011, 0.068]**
AI Usage	Family SES	0.001	0.001	0.573	0.567	0.008	[-0.002, 0.004]
<i>Predicting Digital Literacy</i>							
Digital Literacy	Openness	0.168	0.017	9.801	<0.001	0.179	[0.134, 0.202]***
Digital Literacy	Extraversion	0.191	0.018	10.501	<0.001	0.189	[0.155, 0.226]***
Digital Literacy	Perseverance	0.060	0.016	3.760	<0.001	0.066	[0.029, 0.091]***
Digital Literacy	Family SES	-0.007	0.001	-5.069	<0.001	-0.072	[-0.009, -0.004]***
<i>Predicting Academic Performance</i>							
Test Score	Openness	2.889	0.421	6.864	<0.001	0.096	[2.064, 3.714]***
Test Score	Extraversion	-0.619	0.440	-1.406	0.160	-0.019	[-1.481, 0.244]
Test Score	Perseverance	4.825	0.387	12.461	<0.001	0.166	[4.066, 5.584]***
Test Score	Family SES	0.442	0.034	13.157	<0.001	0.151	[0.376, 0.508]***
Test Score	AI Usage	0.100	0.355	0.283	0.777	0.004	[-0.594, 0.795]
Test Score	Digital Literacy	1.737	0.480	3.617	<0.001	0.054	[0.796, 2.679]***

Note: Raw Est. = unstandardized coefficient; β (Stand.) = standardized coefficient; CI = confidence interval. Control variables included but not shown for brevity. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

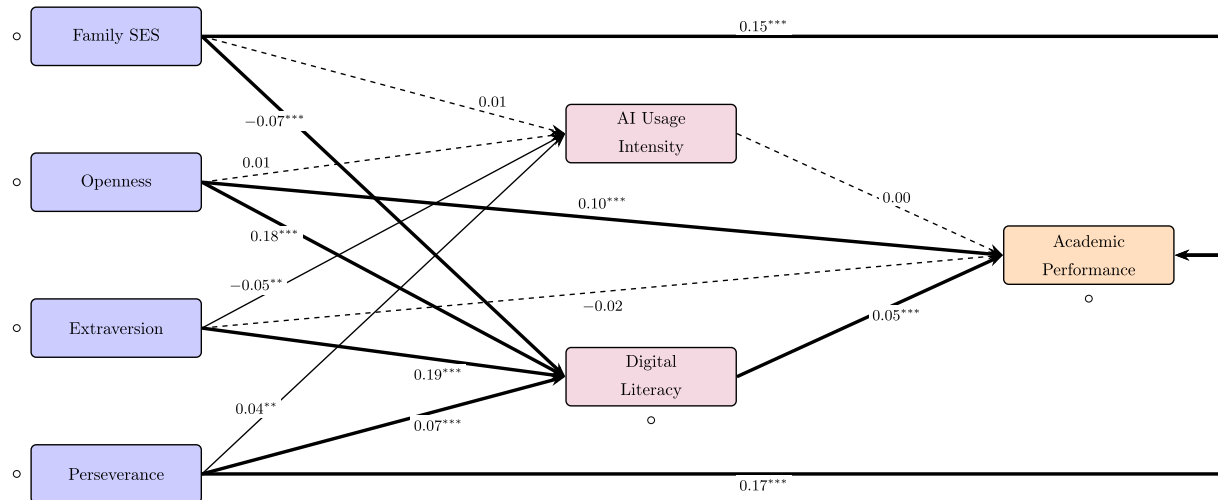


Fig. C.9. SEM Path Diagram: Standardized Coefficients and Significance Levels for Mediation Model. Small circles on the left of independent variables and at the bottom of mediators and dependent variable indicate measurement or structural error terms for latent constructs. Path coefficients were standardized with significance indicators (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Bold lines indicate paths that are part of significant mediation pathways, solid lines indicate other significant paths, and dashed lines indicate non-significant relationships.

The SEM robustness check validates these conclusions under simultaneous pathway estimation rather than sequential single-mediator models. Despite modest model fit indices ($CFI = 0.840$, $TLI = 0.779$) reflecting the complexity of modeling multiple constructs simultaneously, the structural path coefficients demonstrate consistency with PROCESS results: digital literacy maintains its predictive relationship with academic performance ($B = 1.737$, $p < 0.001$), while AI usage shows no significant effects ($p = 0.777$). Because all robustness checks employ identical analytical frameworks, control variables, and bootstrap procedures, observed differences reflect substantive rather than methodological variation. Taken together, these results demonstrate that the differential mediation roles of digital literacy versus AI usage intensity are robust to alternative operationalizations and represent genuine patterns rather than measurement artifacts.

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